Shared iPad for Frontline Teams — Managed Apple IDs, ABM & Microsoft 365 Session Isolation

Implementation Solution Guide

Version 1.0.0 · 2026-07-14

Contents

Executive Summary — Shared iPad for Frontline Teams 5

The situation . . . 5

The root cause . . . 5

The decision . . . 5

The real blocker — and it is not technical . . . 6

What this solution delivers . . . 6

The outcome . . . 6

Honest limitations . . . 7

1. Solution Overview — Shared iPad for Frontline Teams: Managed Apple IDs, ABM & Microsoft 365 Session Isolation 7

The problem this solves . . . 7

The root cause — stated plainly . . . 7

The two supported architectures — the decision you must make . . . 8

How this solution works . . . 10

The Managed Apple ID reclamation workstream — the real blocker . . . 10

Benefits . . . 11

Limitations — be honest with the buyer . . . 11

The standard deployment model . . . 12

What this solution covers, and what the catalog already covers . . . 12

2. Requirements — Shared iPad for Frontline Teams 13

R1. Microsoft Intune licensing and an Intune-managed tenant . . . 13

R2. Microsoft 365 licensing for the field workload . . . 13

R3. Microsoft Entra ID tenant with the frontline staff present . . . 14

R4. Apple Business Manager (ABM) account . . . 14

R5. Verified domain(s) in Apple Business Manager . . . 14

R6. iPads present in Apple Business Manager (this constrains an existing fleet) . . . 14

R7. Apple MDM Push certificate (APNs) . . . 15

R8. Automated Device Enrollment (ADE) token connected to Intune . . . 15

R9. Supervision . . . 15

R10. A supported iPadOS version on every device . . . 15

R11. Device storage sized for Shared iPad's per-user data . . . 16

R12. Managed Apple IDs available for every Shared iPad user . . . 16

R13. ABM federated to Microsoft Entra ID *(strongly recommended)* . . . 16

R14. Personal Apple ID conflicts resolved — **this is a prerequisite, not a step** . . . 16

R15. Administrative roles . . . 17

R16. Network access to Apple and Microsoft endpoints . . . 17
R17. Wi-Fi coverage across the field working area . . . 18
R18. Executive sponsorship and a communications plan for the reclamation campaign . . . 18
Requirements checklist . . . 18
Global Requirements — Intune Solutions (category-level) 19
G1. Microsoft Entra ID tenant . . . 19
G2. Microsoft Intune subscription . . . 19
G3. Administrative roles . . . 19
G4. Device enrollment in Intune . . . 19
G5. Pilot security group . . . 20
G6. Supported, current operating system . . . 20
G7. Network access to required cloud endpoints . . . 20
3. Implementation — Shared iPad for Frontline Teams 20
Phase 1 — Assess & prepare identity . . . 21
Phase 2 — Configure the Apple connection . . . 24
Phase 3 — Configure the device . . . 25
Phase 4 — Pilot . . . 29
Phase 5 — Expand . . . 30
5. Testing & Validation — Shared iPad for Frontline Teams 30
Test 1 — Tenant and Apple readiness . . . 31
Test 2 — Enrollment-time grouping owner . . . 31
Test 3 — The pilot iPad is actually in Shared iPad mode . . . 31
Test 4 — Managed Apple ID sign-in works . . . 32
Test 5 — **THE SESSION-ISOLATION A/B TEST** *(the acceptance gate)* . . . 32
Test 6 — Field lockdown is enforced . . . 34
Test 7 — Compliance reports correctly . . . 35
Test 8 — The field workload actually works . . . 35
Test 9 — Storage holds under multiple resident users . . . 35
Acceptance gate . . . 35
Shift-handover runbook — for the field team . . . 36
6. Troubleshooting — Shared iPad for Frontline Teams 36
1. "The next user still sees the prior user's mail" — **THE #1 CAUSE** . . . 36
2. Managed Apple ID sign-in fails . . . 38
3. ABM says "address already in use" / the Managed Apple ID cannot be created . . . 39
4. Federation sign-in loop, or the Microsoft sign-in page never appears . . . 40
5. Restrictions not applying / the App Store is still there . . . 40

6. The device won't ADE-enroll / the serial isn't on the token . . . 41
7. Storage full / a user can't sign in for lack of space . . . 43
8. The device group is empty after enrollment / nothing deploys . . . 43
9. An app isn't available or misbehaves under Shared iPad . . . 44
10. Compliance shows Not evaluated / the device is Not compliant . . . 45
11. The scripts . . . 46
7. Best Practices — Shared iPad for Frontline Teams 47
Architecture . . . 48
The Apple ID reclamation campaign . . . 48
Devices and procurement . . . 49
Configuration . . . 49
Naming and hygiene . . . 50
Operations . . . 50
Scripts . . . 51
8. Deliverables Summary — Shared iPad for Frontline Teams 51
Documentation (this guide) . . . 51
PowerShell scripts (Microsoft Graph) — `SHAREDIPAD-Scripts.zip` . . . 52
Objects this solution creates . . . 53
Hero image . . . 53
How it's delivered . . . 53
A note on vendor specifics . . . 53
Appendix A — PowerShell Scripts 54

Executive Summary — Shared iPad for Frontline Teams

The situation

A frontline field team runs a pool of iPads — typically 30–50 devices — shared across shifts rather than assigned one-to-one. Staff take whichever iPad is on the rack, use it for work orders, drawings, photos, and Microsoft 365 collaboration, and return it for the next shift.

The devices are locked down. They are supervised. No App Store, no personal apps. And they still fail in the way that matters most:

> ***The next user opens Outlook and sees the previous user's mail.***

Teams keeps the prior chat. The day shift's photos are in the night shift's camera roll. On a device used by a dozen people a week, this is a data-exposure problem — one that is visible to staff, obvious to an auditor, and impossible to explain away.

The root cause

A single-user iPad passed between shifts has no per-user session boundary. The operating system was never told the user changed, so Microsoft 365 apps keep the previous user's cached tokens and data. Signing out of one app does not sign out of the others, and app-level sign-out never clears on-device cached data.

"Tell staff to sign out" is not a fix. It is a procedure that depends on a tired person performing an incomplete action — and even performed perfectly, it does not produce data separation. Adding more restrictions will not help either: **lockdown is not isolation**.

The fix is a real multi-user architecture. There are exactly two supported ones, and the current deployment is neither.

The decision

	Apple Shared iPad *(recommended)*	**Microsoft 365 Shared Device Mode**
Isolation	**OS-level** — each user gets their own data space	**App-level** — participating apps only
Photos, files, browser separated	**Yes**	**No**
Requires Managed Apple IDs + ABM + wipe	**Yes**	No
Fixes the reported problem	**Yes**	**Partly**

Apple Shared iPad is the right answer for shift-based devices, and it is what this solution deploys. Each staff member signs in to the *device* with a **Managed Apple ID** — federated to Microsoft Entra ID so they use their normal work credentials — and iPadOS gives each user their own data space. Sign-out genuinely ends the session. The next user gets a clean device, by architecture rather than by procedure.

Microsoft 365 Shared Device Mode is the fallback where Shared iPad is not viable. It is honest about what it delivers: app-level sign-out across participating Microsoft apps, with no OS-level separation of photos, files, or browser state.

The real blocker — and it is not technical

Over the years, staff created **personal Apple IDs using their work email address**. **Apple Business Manager cannot issue a Managed Apple ID at an address already in use by a personal Apple ID.** Every affected staff member is a blocked Shared iPad user until that conflict is resolved.

This is the constraint that determines your timeline, and it reaches beyond the iPads — a staff member's personal Apple ID carries their personal iPhone, their home Mac, their photo library, and their App Store purchases. Resolving it touches their personal life.

It is therefore an identity, HR, legal, and communications decision — not a technical one. This solution surfaces the exposure, presents the options and their impact, and sequences the campaign. The organization's HR, legal, and identity owners make the call. Executive sponsorship and a real communications plan are prerequisites, not nice-to-haves.

There is a second constraint worth knowing on day one: **iPads not present in Apple Business Manager cannot be enrolled through Automated Device Enrollment, and therefore cannot run Shared iPad at all.** On a fleet accumulated over several years, this can turn a configuration project into a procurement conversation. This solution establishes that number in the first phase, before any date is promised.

What this solution delivers

- **The architecture decision**, framed with honest trade-offs, so the organization chooses knowingly.
- **Click-by-click implementation** — ABM setup, conflict resolution as an explicit gate, Entra

federation, the ADE enrollment profile with Shared iPad enabled, supervision, field lockdown, compliance, the app allowlist, and staged wipe-and-re-enroll of the existing fleet.

- **Five PowerShell scripts** — four read-only (readiness discovery, PASS/FAIL validation, fleet

reporting, and the Apple ID reclamation worklist), and one state-changing helper that **defaults to** `-WhatIf`.

- **The validation gate** — an explicit, reproducible **session-isolation A/B test** with exact

PASS/FAIL criteria: User A signs in, works, signs out; User B signs in on the same device and must see no mailbox, no cached mail, no chat, no photos, no autofill, and no signed-in state belonging to A. Plus a diagnostic table mapping every failure back to the architecture step that caused it.

- **A shift-handover runbook** for the field team and a **staged fleet plan** for ~40 devices.

The outcome

The leak stops. The next user gets a clean device — because the operating system separates them, not because someone remembered to sign out. Staff sign in with their normal work credentials. Any iPad works for any person. The device stays locked to its job.

And critically: **you can prove it.** The session-isolation test is the acceptance gate. It either passes or it does not, and this solution tells you exactly which architecture step to fix when it does not.

Honest limitations

- **Shared iPad requires ABM, ADE, and supervision. There is no shortcut** — no profile or restriction delivers OS-level multi-user on a non-ADE iPad.
- **Existing devices must be wiped and re-enrolled.** Budget the staging time and the shift downtime.
- **Devices not in ABM are out of scope** until they are brought in or replaced.
- **The Apple ID reclamation campaign is the long pole**, and it is a people project.
- **Storage sizing is a real constraint** — undersized iPads will fail sign-ins at scale.
- **Some apps behave differently under Shared iPad** — pilot the CMMS and viewer apps before committing the fleet.
- **Apple's and Microsoft's specifics change.** Current iPadOS floors, storage and user-count guidance,

ABM navigation and conflict-resolution mechanics, federation specifics, and the Microsoft SDM supported-app list are all flagged in this guide as **verify against current vendor documentation** — and are deliberately not asserted here. A guide that invents them is a guide that fails at step 12.

1. Solution Overview — Shared iPad for Frontline Teams: Managed Apple IDs, ABM & Microsoft 365 Session Isolation

The problem this solves

A frontline/field team runs a pool of iPads — typically 30–50 devices — that are **shared across shifts**, not assigned one-to-one. Staff pick up whichever iPad is on the rack, use it for the shift (a CMMS/work-order app, viewing schematics and drawings, taking photos, light Microsoft 365 collaboration), put it back, and someone else picks it up on the next shift.

The deployment is locked down and supervised — no App Store, no personal apps — and it still fails in the one way that matters:

> ***The next user opens Outlook and sees the previous user's mail.***

Teams keeps the prior chat. Photos taken by the day shift are in the camera roll of the night shift. Sign-in fields autofill someone else's identity. On a device used by a dozen people a week, that is a data-exposure problem, not an inconvenience.

The root cause — stated plainly

A supervised, single-user iPad that is passed between shifts has no per-user session boundary.

iPadOS was configured for one owner. Handing the hardware to a different human does not tell the operating system that the user changed — as far as the OS is concerned, every shift is the *same user*. So:

- **Cached tokens and data stay put.** Microsoft 365 apps keep the refresh tokens, mailbox cache,

chat history, and downloaded files of whoever signed in last, because nothing ever ended that user's session.

- **Signing out of one app does not sign out of the others.** Each Microsoft app holds its own session state. Outlook sign-out leaves Teams, OneDrive, and Word signed in.
- **App-level sign-out never removes on-device cached data.** Even a clean sign-out inside an app does not guarantee the local cache, attachments, or photo library are gone.
- **"Tell staff to sign out at the end of the shift" is not a fix.** It is a procedure that depends on a tired person performing an incomplete action, and even performed perfectly it does not deliver data separation. It is not an architecture.

The fix is a real multi-user architecture. There are exactly two supported ones. The deployment described above is neither of them — it is a single-user device being used by many people, which is why it leaks. Adding more restrictions to that profile will never fix it, because lockdown is not isolation.

The two supported architectures — the decision you must make

*Option A — Apple Shared iPad *(recommended for shift-based use, and what this solution deploys)**

Shared iPad is Apple's native, OS-level multi-user feature for iPad. Each user signs in to the *device* with their own **Managed Apple ID** issued by **Apple Business Manager (ABM)** — ideally federated to Microsoft Entra ID so staff sign in with their normal work credentials. iPadOS gives each Managed Apple ID **its own data space on the device**. Signing out genuinely ends that user's session; the next user signs in to a clean environment.

What it requires:

- The iPad is enrolled through **Automated Device Enrollment (ADE)** via **Apple Business Manager** and is **supervised**. Shared iPad is only available through ADE — you cannot switch an existing manually-enrolled iPad into Shared iPad without wiping and re-enrolling it.
- **Shared iPad is enabled in the ADE enrollment profile** in Intune.
- Every user has a **Managed Apple ID**. This is the hard part — see the reclamation workstream below.

Trade-offs to state honestly:

- **It requires ABM and Managed Apple IDs.** If your staff have personal Apple IDs on their work email address, Managed Apple ID creation is blocked until those conflicts are resolved. That is a people-and-comms project, not a checkbox.
- **Device storage sizing matters.** Each signed-in user consumes space on the device for their own data. More users resident on a device means more storage consumed, and an undersized iPad will refuse sign-ins or evict users. *(Apple publishes current minimum/recommended storage and maximum resident-user guidance — **verify the current figures against Apple's Shared iPad documentation before you size or buy hardware**. Do not assume the numbers from an older deployment still hold.)*
- **Converting an existing fleet means wipe and re-enroll.** Every iPad in the pool gets erased and

re-enrolled through ADE. Plan the staging and the downtime.

• **Some apps and features behave differently under Shared iPad.** Not every app is designed for a multi-user iPad, and some iPadOS features are unavailable or altered in this mode. **Verify your specific app set (especially the CMMS app) against the vendor's documentation and a pilot device before committing the fleet.**

• **Devices must be in ABM.** iPads that were bought retail and never linked to ABM through a reseller or Apple cannot be ADE-enrolled — which means they cannot run Shared iPad. See Requirements R6.

Option B — Microsoft 365 Shared Device Mode (SDM)

Shared Device Mode is Microsoft's app-level feature. The device is marked as shared, and participating Microsoft apps use the **Microsoft Authenticator broker** for a global sign-in/sign-out: signing out of one participating app signs the user out of all of them.

Trade-offs to state honestly:

• **It is app-level, not OS-level.** SDM does **not** give you per-user operating-system data separation. The camera roll, the Files app, Safari state, and anything outside the participating apps are still one shared pool.

• **It only covers apps that participate in SDM.** An app that does not implement shared device mode is unaffected by it and behaves exactly as it does today.

• **The set of Microsoft apps that support shared device mode on iOS/iPadOS changes over time.**

Verify the current supported-app list against Microsoft's documentation before you design around it. This solution does not assert which Microsoft apps do or do not support SDM on iPadOS, because that list moves and a stale list will produce a design that fails validation.

How to choose

	A — Apple Shared iPad	B — Microsoft 365 SDM
Isolation level	**OS-level** — per-user data space	**App-level** — participating apps only
Photos / Files / Safari separated	**Yes**	**No**
Requires Managed Apple IDs	**Yes**	No
Requires ABM + ADE + supervision	**Yes**	No (device just needs to be marked shared)
Requires wipe of existing fleet	**Yes**	No
Fixes "next user sees prior mail"	**Yes**	Partly — only for participating apps

Choose A when the device is genuinely shift-shared, the workload includes photos/files/browser state, and the organization can adopt ABM and Managed Apple IDs. **That is this scenario, and it is what this solution deploys.**

Choose B when Shared iPad is not viable — Managed Apple IDs cannot be adopted, the devices cannot be brought into ABM, or the fleet cannot be wiped — and accept that you are buying a partial fix that covers participating Microsoft apps only.

They are not mutually exclusive. A and B can be complementary: Shared iPad gives the OS boundary, and SDM (where the apps support it) tightens the sign-out behaviour of the Microsoft app suite on top of it. Deploy A first; layer B only after verifying app support.

How this solution works

1. **Apple Business Manager is stood up and verified** — the organization's ABM account, the domains

it will issue Managed Apple IDs on, and the devices (either purchased through a linked reseller or added by Apple's supported method).

1. **The personal-Apple-ID conflicts are resolved — this is a gate, not a step.** Staff who created a

personal Apple ID using their **work email address** block Managed Apple ID creation on that address. Nothing downstream works until this campaign runs. See below.

1. **ABM is federated to Microsoft Entra ID** so a Managed Apple ID signs in with the staff member's

normal work credentials rather than a second password to remember.

1. **An ADE enrollment profile with Shared iPad enabled** (`SHARED IPAD – ADE Profile – Shared`) is

created in Intune and assigned to the iPad serial numbers on the ADE token.

1. **The field lockdown profile** (`SHARED IPAD – Restrictions – Field Lockdown`) strips the device to

its job: no App Store, no personal apps, no adding personal accounts.

1. **The field app allowlist** (`SHARED IPAD – Apps – Field Allowlist`) deploys and exposes only the

CMMS app, the schematics/document viewer, Camera, and the limited Microsoft 365 set the shift actually needs.

1. **A compliance policy** (`SHARED IPAD – Compliance – iPadOS`) reports the health of the fleet.
2. **The existing iPads are wiped and re-enrolled through ADE** in staged waves.
3. **The session-isolation test is run** — User A signs in, uses the device, signs out; User B signs

in and must see **nothing** of A's. That test is the acceptance gate for the whole project.

The Managed Apple ID reclamation workstream — the real blocker

This is the part that sinks projects, and it is not technical.

Over the years, staff created **personal Apple IDs using their organization email address** — to buy apps, set up a personal iPhone, use iCloud. **Apple Business Manager cannot issue a Managed Apple ID at an address that is already in use by a personal Apple ID.** Every affected staff member is a blocked Shared iPad user until their conflict is resolved.

What you need to understand about it:

- **It reaches every Apple device that person owns**, not just the work iPad. Their personal iPhone,

their home Mac, their iCloud photo library, their App Store purchase history — all of it is tied to that Apple ID. Resolving the conflict touches their personal life. This needs executive sponsorship and a genuine communications plan, not a technical push and a hope.

- **Apple provides a supported conflict-resolution path** built around verifying ownership of your domain, after which affected personal Apple IDs must be changed to a personal email address (or are otherwise handled by Apple's process). **The exact flow names, portal navigation, notification timelines, and mechanics are Apple's and they change — verify every specific against current Apple Business Manager documentation before you plan dates or write user comms.** This solution gives you the concept, the decision points, and the sequencing; Apple's documentation gives you the clicks.

- **The scripts in this package build the *candidate* list, not the authoritative one.**

`SHAREDIPAD-AppleIDConflict.ps1` reads Entra ID and tells you which staff have work addresses that would need a Managed Apple ID and therefore *may* collide. **Only Apple Business Manager knows which addresses actually collide.** Use the script's CSV to plan and track the campaign; use ABM for truth.

This is a decision, not a configuration. Whether to require staff to rename or surrender a personal Apple ID that carries their photos and purchases is an **identity, HR, legal, and communications decision**. This solution presents the options and their impact. **The customer's HR, legal, and identity owners make the call — never make it for them, and never let the technical team make it by default.**

Benefits

- **The leak stops.** Per-user OS data separation means the next user gets a clean device — not a procedure that hopes they signed out.

- **Sign-out actually means sign-out.** Ending a Shared iPad session ends the session, across every app, at the OS layer.

- **Staff use their normal work credentials** when ABM is federated to Entra ID — no second password.

- **The fleet is interchangeable.** Any staff member can pick up any iPad and get their own environment.

- **The device stays locked to its job** — no App Store, no personal accounts, only the field workload.

- **It is auditable.** The included scripts report Shared iPad readiness and state per device across the fleet.

Limitations — be honest with the buyer

- **Shared iPad requires ABM + ADE + supervision. There is no shortcut.** No configuration profile, no restriction, and no app setting delivers OS-level multi-user on a non-ADE iPad.

- **Existing devices must be wiped and re-enrolled.** Budget the staging time and the shift downtime.

- **Devices not in ABM cannot be ADE-enrolled** — and therefore cannot run Shared iPad at all. If part

of your fleet was bought retail, that part of the fleet is out of scope until it is brought into ABM (or replaced).

• **The Apple ID reclamation campaign is the long pole**, and it is a people project with a technical tail.

• **Storage sizing is a real constraint.** Under-specced iPads will run out of room for resident users. *(Verify Apple's current guidance — see Requirements R8.)*

• **Some apps behave differently under Shared iPad.** Pilot your actual CMMS and viewer apps before committing the fleet.

• **This solution does not assert the Microsoft SDM iOS app-support list** — verify it against Microsoft's current documentation if you layer Option B.

The standard deployment model

Every step in this solution follows the standard pattern:

Assess → Configure → Pilot → Validate → Expand.

• **Assess** — ABM state, ADE token, device eligibility, storage, and the Apple ID conflict exposure.

• **Configure** — federation, ADE profile with Shared iPad, lockdown, compliance, app allowlist.

• **Pilot** — one iPad, two real users (`PILOT – Shared iPad Users`), `SHARED IPAD – FMD – Pilot`.

• **Validate** — the session-isolation A/B test with explicit PASS/FAIL criteria (Testing §5).

• **Expand** — staged waves into `SHARED IPAD – FMD – Production`.

What this solution covers, and what the catalog already covers

This solution is deliberately scoped to the gap. It **cross-references** — and does **not** reproduce — three existing catalog solutions:

Existing solution	What it covers	Why it is not this
Implement iOS Lock Down Profile for iOS iPhones in Microsoft Intune for Shared Devices	Supervised **single-user iPhone** lockdown — restrictions and kiosk-style hardening on a pooled handset.	It hardens a **single-user** device. It contains no multi-user architecture, so it cannot separate sessions. Use it for iPhone lockdown; use *this* for iPad session isolation.
iOS iPhone Shared Device Mode (Lockdown) with WebEx Calling in Microsoft Intune	Microsoft **app-level** shared device mode on iPhone, plus kiosk lockdown to a calling app.	That is **Option B** applied to iPhone. It is app-level, not OS-level, and it is not Shared iPad.

Existing solution	What it covers	Why it is not this
Implement Entra ID Integration with ABM for Account Auto Provisioning for Microsoft Intune	**ABM account auto-provisioning** from Entra ID — getting accounts into ABM.	It provisions accounts into ABM. It does **not** cover Shared iPad, Managed Apple ID conflict resolution, or session isolation. It is a useful companion once ABM is stood up.

This solution covers: Apple Shared iPad (OS-level multi-user) + Managed Apple IDs + the personal Apple ID reclamation workstream + field lockdown + the session-isolation validation gate. That is net-new — no existing catalog solution delivers per-user session isolation on iPad.

2. Requirements — Shared iPad for Frontline Teams

Confirm **every** requirement below before you begin. Each entry follows the same pattern: **Description · Why required · How to verify · How to implement if missing.**

Category-wide prerequisites (Entra tenant, Intune licensing, admin roles, pilot group, supported OS, network) are covered once in the **Global Requirements — Intune Solutions** document. The items here are specific to **Apple Shared iPad with Managed Apple IDs**.

> ***Verification discipline.*** *Several requirements below depend on* ***Apple-published specifics*** *(iPadOS version floors, storage minimums, maximum resident users, ABM portal navigation, federation and conflict-resolution mechanics). Those change. Where you see a* ***VERIFY WITH APPLE*** *flag, go to current Apple Business Manager / Shared iPad documentation and confirm the figure or flow* ***before*** *you buy hardware, publish a date, or write user communications. Do not carry a number forward from an older project.*

R1. Microsoft Intune licensing and an Intune-managed tenant

Description: An Intune-bearing licence plan (Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1) covering every staff member who will sign in to a Shared iPad. **Why required:** Intune delivers the ADE enrollment profile that enables Shared iPad, plus the restrictions, compliance, and app assignments. **How to verify:** Sign in to `https://intune.microsoft.com` → **Tenant administration** → **Tenant status**; confirm an active subscription. Then Entra → **Users** → **(a pilot user)** → **Licenses**; confirm an Intune-bearing licence. **How to implement if missing:** Purchase and assign the licence to the pilot users, then to the full frontline population before Expand.

R2. Microsoft 365 licensing for the field workload

Description: Microsoft 365 licences covering the mail/collaboration apps the shift actually uses (e.g. Exchange Online for Outlook, Teams). **Why required:** The session-isolation test signs a user in to Outlook and Teams and requires a working mailbox and chat. An unlicensed pilot user produces a false FAIL. **How to verify:** Entra → **Users** → **(pilot user)** → **Licenses**; confirm the service plans for the apps in the allowlist are **On**. **How to implement if missing:** Assign the appropriate Microsoft 365 licence to the pilot users before Testing §5.

R3. Microsoft Entra ID tenant with the frontline staff present

Description: An Entra ID tenant containing accounts for every staff member who will use a Shared iPad. **Why required:** Federating ABM to Entra ID means the Managed Apple ID signs in with the Entra account. No Entra account, no federated sign-in. It is also the source for the Apple ID conflict-candidate list (`SHAREDIPAD-AppleIDConflict.ps1`). **How to verify:** Entra → **Users**; confirm the frontline staff exist with correct, current UPNs and mail addresses on the domain you intend to federate. **How to implement if missing:** Create or synchronise the accounts before starting the reclamation campaign — the campaign is built from this list.

R4. Apple Business Manager (ABM) account

Description: An Apple Business Manager account for the organization, at `https://business.apple.com`. **Why required:** ABM is what issues **Managed Apple IDs** and what holds the **ADE token**. Both are mandatory for Shared iPad. There is no path to Shared iPad without ABM. **How to verify:** Sign in to `https://business.apple.com` with an ABM administrator account and confirm the organization is enrolled and active. **How to implement if missing:** Enrol the organization in Apple Business Manager. **VERIFY WITH APPLE** — the enrolment process, the roles required, and the verification Apple performs are Apple's and change; follow Apple's current ABM documentation. This is typically a multi-day process involving an organizational verification contact, so start it early.

R5. Verified domain(s) in Apple Business Manager

Description: The email domain(s) on which you will issue Managed Apple IDs (e.g. `contoso.com`), verified in ABM. **Why required:** ABM will only issue Managed Apple IDs on a domain the organization has proven it owns. Domain verification is also the entry point to Apple's **personal Apple ID conflict-resolution** path (R11). **How to verify:** In ABM, review the organization's domain settings and confirm the intended domain shows as verified. **How to implement if missing:** Verify the domain in ABM. **VERIFY WITH APPLE** — the exact ABM navigation, the DNS record type/value Apple asks you to publish, and the timing are Apple's and change. Follow Apple's current ABM documentation, then publish the record in your DNS (Cloudflare, Route 53, or wherever the zone lives).

R6. iPads present in Apple Business Manager (this constrains an existing fleet)

Description: Every iPad that will run Shared iPad must appear in **ABM** → **Devices**, either because it was purchased through a reseller linked to your ABM organization, or because it was added by an Apple-supported method. **Why required: Shared iPad is only available on devices enrolled via Automated Device Enrollment (ADE), and ADE is only available for devices that are in ABM.** This is the single hardest constraint on an existing fleet.

> ***Read this before you plan.*** *If your ~40 iPads were bought retail — from a store, from a non-linked online seller, or as a mixed bag over several years —* ***they are not in ABM and cannot be ADE-enrolled****, which means* ***they cannot run Shared iPad at all****, no matter what you configure in Intune. There is no Intune setting that fixes this. Your options are to bring them into ABM by an Apple-supported route, or to replace the ineligible devices with iPads purchased through an ABM-linked reseller.* ***VERIFY WITH APPLE*** *— which routes exist for adding already-owned devices to ABM, the eligibility rules, and any provisional/opt-out window are Apple's and change. Establish this*

> ***before*** *you promise the customer a date, because it determines whether the project is a configuration exercise or a hardware purchase.*

How to verify: In ABM → **Devices**, search for the serial numbers of your existing iPads. Serials that return no result are not in ABM. Run `SHAREDIPAD-Discovery.ps1` to export the Intune-side serial inventory to compare against. **How to implement if missing:** Link your reseller to ABM and have them add the devices they sold you; or bring the devices in by Apple's current supported method; or purchase replacement iPads through an ABM-linked reseller. **VERIFY WITH APPLE.**

R7. Apple MDM Push certificate (APNs)

Description: A valid **Apple MDM Push certificate** uploaded to Intune. **Why required:** Every iOS/iPadOS enrollment and all ongoing MDM communication runs over APNs. An expired APNs certificate silently breaks the entire fleet. **How to verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate**; status shows **Active** with a future expiry. `SHAREDIPAD-Discovery.ps1` reports the expiry date. **How to implement if missing:** Create it (Implementation §Step 5) using a **company-owned Apple ID** that you record and reuse every year. **Never use a staff member's personal Apple ID** — when they leave, the certificate cannot be renewed and every managed device must be re-enrolled.

R8. Automated Device Enrollment (ADE) token connected to Intune

Description: An **enrollment program token** downloaded from ABM and uploaded to Intune, linking the two. **Why required:** The ADE token is how Intune sees your ABM devices and how it delivers the enrollment profile that **enables Shared iPad**. No token, no ADE, no Shared iPad. **How to verify:** Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens**; a token is listed with a **future** expiry, and opening it shows your iPad serials under **Devices**. `SHAREDIPAD-Discovery.ps1` reports token presence and expiry. **How to implement if missing:** Create the MDM server in ABM and upload the token to Intune (Implementation §Step 6). **The token expires annually — put the renewal in the operational calendar.** An expired token stops new enrollments dead.

R9. Supervision

Description: The iPads must be **supervised**. **Why required:** Supervision is what unlocks the deeper iPadOS restrictions the field lockdown relies on (blocking the App Store, blocking account modification, blocking app install/removal). ADE enrollment supervises the device. **How to verify:** Intune → **Devices** → **All devices** → **(device)** → **Properties**; **Supervised = Yes**. `SHAREDIPAD-Discovery.ps1` reports supervision per device. **How to implement if missing:** Enrol through ADE with **Supervised** enabled in the enrollment profile (Implementation §Step 7). A device that is not supervised must be wiped and re-enrolled.

R10. A supported iPadOS version on every device

Description: A current, Apple-supported iPadOS release on each iPad, and hardware that Apple lists as Shared iPad-capable. **Why required:** Shared iPad has a minimum iPadOS version and a supported-hardware list. An iPad below the floor, or a model Apple does not list, will not run Shared iPad regardless of your enrollment profile. **How to verify:** On the device, **Settings** → **General** → **About**; or Intune → **Devices** → **All devices** (the OS version column). `SHAREDIPAD-Discovery.ps1` exports model + iPadOS version per device. **How to implement if missing:** Update iPadOS before enrollment

and set the floor in `SHARED IPAD – Compliance – iPadOS`. **VERIFY WITH APPLE** — the current minimum iPadOS version for Shared iPad and the supported iPad models are Apple's and change with each release. Check Apple's current Shared iPad documentation; do not assume an older floor still applies.

R11. Device storage sized for Shared iPad's per-user data

Description: Enough on-device storage for the number of users you expect to be resident on each iPad simultaneously. **Why required:** Under Shared iPad, **each signed-in user consumes storage on the device** for their own data space. This is the requirement teams get wrong. Buy iPads sized for one user and put twelve shift workers on them and you will get sign-in failures, aggressive eviction of users' data, and a support queue. **How to verify:** Intune → **Devices** → **(device)** → **Hardware** shows total and free storage. Compare against Apple's published guidance for your intended user count. **How to implement if missing:** Size or replace the hardware. **VERIFY WITH APPLE** — Apple publishes current **minimum and recommended storage** for Shared iPad and a **maximum number of resident users**; both have changed across iPadOS releases. **Get these figures from Apple's current Shared iPad documentation before you specify or purchase hardware.** Getting this wrong is a hardware purchase, not a reconfiguration.

R12. Managed Apple IDs available for every Shared iPad user

Description: A **Managed Apple ID**, issued by ABM, for every staff member who will sign in to a Shared iPad. **Why required: Shared iPad sign-in is by Managed Apple ID.** It is the identity the OS uses to create and separate the per-user data space. No Managed Apple ID, no Shared iPad session — the user simply cannot sign in. **How to verify:** In ABM → **Users**, confirm the pilot users hold Managed Apple IDs on your verified domain. Run `SHAREDIPAD-AppleIDConflict.ps1` to produce the Entra-side candidate list to reconcile against. **How to implement if missing:** Create Managed Apple IDs in ABM — manually, by federation (R13), or by the account auto-provisioning covered in the catalog's *Implement Entra ID Integration with ABM for Account Auto Provisioning for Microsoft Intune* solution. **R14 is a gate on this** — a conflicted address will not issue.

R13. ABM federated to Microsoft Entra ID *(strongly recommended)*

Description: ABM federation configured against Microsoft Entra ID, so Managed Apple ID sign-in authenticates against the staff member's normal Entra credentials. **Why required:** Without federation, every staff member gets a **second, separate password** for their Managed Apple ID. On a frontline team that shares devices across shifts, a second password is a support burden and a security regression (it gets written on the rack). Federation also means disabling the Entra account disables Shared iPad access. **How to verify:** In ABM, review the organization's federated-authentication settings and confirm the Entra ID connection is active for your verified domain; then sign a pilot user in to a Shared iPad and confirm they are prompted by the Microsoft sign-in page. **How to implement if missing:** Configure ABM federation to Entra ID (Implementation §Step 4). **VERIFY WITH APPLE AND MICROSOFT** — the ABM federation flow, the ABM navigation, the Entra consent/permissions it requests, the behaviour for accounts that already exist, and the handling of domains with existing Apple IDs are Apple's and Microsoft's, and they change. **Follow Apple's current ABM federation documentation for the exact steps.** This solution covers the concept, the sequencing, and the decision points.

R14. Personal Apple ID conflicts resolved — this is a prerequisite, not a step

Description: No staff member who needs a Managed Apple ID may have a **personal Apple ID** already using that work email address. **Why required: ABM cannot issue a Managed Apple ID at an address that is already in use by a personal Apple ID.** Every conflicted staff member is a blocked Shared iPad user. On a fleet where staff historically tied their org email to a personal Apple ID, this is the largest single blocker in the project, and it is not solved in Intune. **How to verify:**

1. Run `SHAREDIPAD-AppleIDConflict.ps1` to build the **candidate** worklist from Entra ID — every user

whose work address would need a Managed Apple ID and therefore *may* collide. **The script cannot query Apple.** It produces a planning and tracking list.

1. Get the **authoritative** list from **Apple Business Manager** — ABM is the only source that knows

which addresses actually conflict. Reconcile the two. **How to implement if missing:** Run the reclamation campaign (Implementation §Step 3). Apple provides a supported conflict-resolution path built around domain verification, in which affected personal Apple IDs must be changed to a personal email address (or are otherwise captured by Apple's process). **VERIFY WITH APPLE** — the flow names, the ABM navigation, the notification wording Apple sends, the grace period Apple allows, and the outcome for users who do not act are **all Apple's and all change**. **Get them from current Apple Business Manager documentation before you communicate any date to staff.**

> ***Scope warning — this reaches beyond the iPads.*** *A staff member's personal Apple ID carries their personal iPhone, their home Mac, their iCloud photo library, and their App Store purchases. Resolving the conflict touches all of it. This requires executive sponsorship and a real communications plan with a grace period — not a technical push.*
>
> ***This is a decision for the customer, not for you.*** *Whether to require staff to rename or surrender a personal Apple ID tied to their work address is an* ***identity, HR, legal, and communications decision****. Present the options and the impact to the customer's* ***HR, legal, and identity owners*** *and let them decide. Do not make the call, and do not let it be made by default by the person running the script.*

R15. Administrative roles

Description: Intune Administrator (or Global Administrator for connector/federation setup) in Microsoft, **plus** an **Administrator** role in Apple Business Manager. **Why required:** The ADE token exchange, federation, and enrollment profile creation all require elevated rights on **both** sides. A project stalls fast when the person with Intune rights has no ABM account. **How to verify:** Entra → **Roles and administrators** → confirm **Intune Administrator**. In ABM → review the account's role. **How to implement if missing:** Assign **Intune Administrator** in Entra; have the ABM organization owner grant the appropriate ABM role. **Identify a named ABM administrator on the customer side before you start** — this is a common project stall.

R16. Network access to Apple and Microsoft endpoints

Description: Outbound HTTPS from the field network to **Apple's** device-management, activation, and Managed Apple ID sign-in endpoints **and** Microsoft's Entra/Intune/Microsoft 365 service URLs. **Why required:** ADE enrollment contacts Apple at Setup Assistant; Managed Apple ID sign-in and per-user data setup contact Apple on every shift change; policy sync and Microsoft 365 contact Microsoft. A proxy that blocks Apple's endpoints produces an iPad that will not enrol and sign-ins that hang — and it looks exactly like a policy problem. **How to verify:** From a device on the field Wi-Fi, reach

`https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. For Apple, confirm the endpoints Apple publishes for enterprise networks are reachable. **How to implement if missing:** Allow Microsoft's published Office 365/Intune URL & IP ranges and Apple's published enterprise network requirements through the proxy/firewall. **VERIFY WITH APPLE** — Apple's published network requirements list for enterprise/education device management changes; use Apple's current list rather than an older allowlist.

R17. Wi-Fi coverage across the field working area

Description: Reliable Wi-Fi wherever a shift starts and ends. **Why required:** A Shared iPad user sign-in and sign-out are **network operations** — the device authenticates and syncs the user's data space. A user signing in at a dead spot in the yard gets a failure that looks like a Managed Apple ID problem. **How to verify:** Walk the shift-handover locations with a device and confirm signal and successful sign-in. **How to implement if missing:** Fix coverage at the handover points at minimum, or designate a handover station with known-good coverage and put it in the shift runbook (Testing §5).

R18. Executive sponsorship and a communications plan for the reclamation campaign

Description: A named executive sponsor and an agreed communications plan before the Apple ID reclamation campaign starts. **Why required:** You are about to ask staff to change something on their **personal** devices. Without sponsorship and clear comms, the campaign produces a helpdesk flood, a union conversation, or a stalled project — and the technical work sits idle waiting for it. **How to verify:** A named sponsor, agreed messaging, an agreed grace period, and an agreed support path exist in writing before the first notification is sent. **How to implement if missing:** Escalate. **Do not start the reclamation campaign without it.** This is a genuine prerequisite, and it is the one most often skipped.

Requirements checklist

Run `SHAREDIPAD-Discovery.ps1` for the Microsoft-side items; confirm the Apple-side items in ABM.

#	Requirement	Blocking?
R1	Intune licensing	Yes
R2	Microsoft 365 licensing for the field workload	Yes (for the validation test)
R3	Entra ID tenant with frontline staff	Yes
R4	Apple Business Manager account	Yes
R5	Verified domain in ABM	Yes
R6	iPads present in ABM	**Yes — determines feasibility**
R7	Apple MDM Push certificate	Yes
R8	ADE token connected to Intune	Yes
R9	Supervision	Yes

#	Requirement	Blocking?
R10	Supported iPadOS version + model	Yes
R11	Storage sized for per-user data	Yes
R12	Managed Apple IDs for every user	Yes
R13	ABM federated to Entra ID	Strongly recommended
R14	Personal Apple ID conflicts resolved	**Yes — the gate**
R15	Admin roles in Intune *and* ABM	Yes
R16	Network access to Apple + Microsoft	Yes
R17	Wi-Fi at the handover points	Yes
R18	Sponsorship + comms plan	**Yes — for R14**

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level on the marketplace so individual solution listings stay small. Confirm these are in place before deploying *any* Intune solution.

Each entry: **Description · Why required · How to verify · How to implement if missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, enrollment, and security policies. **Verify:** `https://intune.microsoft.com` loads and **Tenant administration** → **Tenant status** shows an active subscription. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies, enrollment connectors, and configuration profiles requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator** (or Global Administrator for connector setup). **Implement:** Assign **Intune Administrator** to the operator; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted devices are enrolled and manageable in Intune. **Why:** Policies only reach enrolled (or app-managed) devices, regardless of platform. **Verify:** Intune → **Devices** → **All devices**; the device

appears and is **compliant**. **Implement:** Configure the appropriate enrollment method for the platform and ownership model, then enroll the device.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted to a small group before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists with intended members. **Implement:** Entra → **Groups** → **New group** → type **Security** → add pilot members.

G6. Supported, current operating system

Description: A supported OS baseline with current updates for the targeted platform. **Why:** Modern Intune features and security settings assume a patched, vendor-supported OS. **Verify:** Check the device's OS version or Intune → device → **Hardware**. **Implement:** Deploy update policies or upgrade to a supported release.

G7. Network access to required cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation — Shared iPad for Frontline Teams

Follow these steps **in order**. The *why* lives in the Solution Overview; the steps below are short and exact so you can **click to implement**.

The design: iPads enrolled through **Automated Device Enrollment (ADE)** from **Apple Business Manager**, **supervised**, with **Shared iPad enabled in the enrollment profile**; each staff member signs in with a **Managed Apple ID** federated to Microsoft Entra ID; the device is locked to the field workload. Pilot first: **Assess → Configure → Pilot → Validate → Expand.**

Consoles used:

- **Microsoft Intune admin center** — `https://intune.microsoft.com`
- **Microsoft Entra admin center** — `https://entra.microsoft.com`
- **Apple Business Manager (ABM)** — `https://business.apple.com`

> ***Apple-portal steps carry a VERIFY WITH APPLE flag.*** *Apple changes ABM's navigation, flow names, and mechanics without notice. Where you see the flag, this guide gives you **what to do and why, and the decision you must make** — get the **exact current clicks** from Apple's live ABM documentation. Do not follow a screenshot from an older project.*

> ***Objects created by this solution** — reuse these names **verbatim** everywhere: | Object | Type | Where | |---|---|---| | `PILOT – Shared iPad Users` | Entra security group (Assigned, **users**) | Step 1 | | `SHARED IPAD – FMD – Pilot` | Entra security group (Assigned, **devices**) | Step 2 | | `SHARED`*

`IPAD – FMD – Production` | *Entra security group (Assigned, **devices**)* | *Step 2* | | `SHARED IPAD – ADE Profile – Shared` | *Intune ADE enrollment profile* | *Step 7* | | `SHARED IPAD – Restrictions – Field Lockdown` | *Intune device configuration profile* | *Step 9* | | `SHARED IPAD – Compliance – iPadOS` | *Intune compliance policy* | *Step 10* | | `SHARED IPAD – Apps – Field Allowlist` | *Intune app assignments (the app set)* | *Step 11* |

Phase 1 — Assess & prepare identity

Step 1. Create the pilot user group

1. Go to `https://entra.microsoft.com` → **Identity** → **Groups** → **All groups** → **New group**.
2. **Group type:** `Security`. **Group name:** `PILOT – Shared iPad Users`.
3. **Membership type:** `Assigned`.
4. **Members** → **No members selected** → add the **two** real staff members who will run the session-isolation test (Testing §5). You need exactly two humans, not one — the test is A-then-B.
1. Click **Create**.

Step 2. Create the device groups

1. Entra → **Identity** → **Groups** → **All groups** → **New group**.
2. **Group type:** `Security`. **Group name:** `SHARED IPAD – FMD – Pilot`.
3. **Membership type:** `Assigned`. Click **Create**.
4. Repeat and create a second group named `SHARED IPAD – FMD – Production`, also `Assigned`.

> ***Both groups must be** `Assigned`**, never** `Dynamic`**.** These groups target devices **at enrollment** (no user affinity is involved at the device layer under Shared iPad), and enrollment-time device grouping only works against an **Assigned** group. Continue to Step 2b.*

Step 2b. Add the Intune Provisioning Client as an owner of each device group

*(Do this for **both** `SHARED IPAD – FMD – Pilot` and `SHARED IPAD – FMD – Production`.)*

1. Entra → **Identity** → **Groups** → **All groups** → open `SHARED IPAD – FMD – Pilot`.
2. Select **Owners** → **Add owners**.
3. In the search box, paste the **AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`. Select the result and click **Select**.

• **Search by the AppId, not by the display name.** The display name varies by tenant — many tenants show it as `Intune Autopilot ConfidentialClient`, others show something else.

• **If the principal does not appear:** open the Intune portal (`https://intune.microsoft.com`) once as a **Global Administrator** to provision it into the tenant, then repeat this step.

1. Confirm the principal is now listed under **Owners**.
2. **Repeat steps 1–4 for** `SHARED IPAD – FMD – Production`**.**

> ***Verify this owner is present before every pilot and every expansion wave.*** *Enrollment-time device grouping works by this principal *owning* the Assigned group. Without the owner, the group stays* ***empty at enrollment*** *and* ***nothing deploys*** *— no restrictions, no compliance, no apps — and the failure looks exactly like a policy problem.*

Step 3. Resolve the personal Apple ID conflicts — THIS IS A GATE

Do not proceed past this step for a given user until their conflict is resolved. A conflicted address cannot be issued a Managed Apple ID, and a user without a Managed Apple ID cannot sign in to a Shared iPad. You can configure everything downstream, but you cannot pilot.

3a. Build the candidate worklist (Microsoft side, read-only).

1. On an admin workstation, run:

```
.\SHAREDIPAD-AppleIDConflict.ps1 -GroupName 'PILOT – Shared iPad Users' -CsvPath .\appleid-candidates.c
```

1. This exports every user whose work address would need a Managed Apple ID and therefore **may** collide with a personal Apple ID. **It cannot query Apple.** It is your planning and tracking list.

1. For the pilot, scope it to `PILOT – Shared iPad Users`. For the campaign, re-run it across the full frontline population.

3b. Get the authoritative conflict list (Apple side).

1. Sign in to `https://business.apple.com` as an ABM administrator.

2. Confirm your domain (e.g. `contoso.com`) is **verified** (Requirements R5). Domain verification is what gives you visibility of, and a path to resolve, personal Apple IDs on your domain.

1. Use ABM's conflict-resolution capability to obtain the list of addresses on your verified domain that are already in use by a **personal** Apple ID.

1. Reconcile against `appleid-candidates.csv` from 3a. **ABM is the truth. The CSV is the tracker.**

> ***VERIFY WITH APPLE.*** *Apple's domain-verification and conflict-resolution capability, its exact name, where it appears in ABM, what it reports, what notification Apple sends the affected user, the* ***grace period*** *Apple allows, and* ***what happens to a user who does not act within it*** *are* ***all Apple's and all subject to change****.* ***Get every one of these specifics from current Apple Business Manager documentation before you plan dates or write a word of user communication.*** *This guide gives you the concept and the sequence, deliberately and on purpose — an invented timeline here would become a broken promise to staff.*

3c. Make the decision — escalate it, do not make it.

The concept is: affected staff must **change their personal Apple ID to a personal email address**, so the work address is released and ABM can issue a Managed Apple ID on it. That request reaches the staff member's **personal iPhone, personal Mac, iCloud photo library, and App Store purchase history**.

This is an identity, HR, legal, and communications decision. It is not yours.

1. Present to the customer's **HR, legal, and identity owners**:

• **Option 1 — Mandate.** All affected staff must move their personal Apple ID to a personal address

within an agreed window. *Impact:* fastest path to Shared iPad; touches personal devices and personal data; needs HR/legal sign-off and probably a consultation process; will generate helpdesk volume and some genuine upset.

• **Option 2 — Voluntary with a deadline.** Staff are asked to move, with support, ahead of an announced date after which Apple's process applies. *Impact:* better received; slower; leaves a tail of non-responders you must still handle.

• **Option 3 — Scope down.** Issue Managed Apple IDs only to staff with no conflict, and defer or exclude the rest. *Impact:* no personal-device impact for the excluded; those staff cannot use a Shared iPad, so the fleet is only partly usable and you carry two operating models.

• **Option 4 — Alternative addressing.** Issue Managed Apple IDs on a **different verified subdomain **the staff member has never used personally (e.g. `staff.contoso.com`). *Impact:* may sidestep many conflicts entirely; adds a second domain to verify and federate, and a second address for staff to understand.** VERIFY WITH APPLE** that this fits your ABM/federation configuration before proposing it as a solution — it is a genuine option worth raising, not a guaranteed escape.

1. **Record their decision in writing.** Then execute it.
2. **Do not send a single staff notification until Requirement R18 is satisfied** — a named executive sponsor, agreed messaging, an agreed grace period, and an agreed support path.

3d. Run the campaign and track it.

1. Send the agreed communication to the affected staff, with the support path.
2. Track each user's state in `appleid-candidates.csv` (the script emits a `Status` column for this).
3. Re-check ABM as staff act; a released address becomes eligible for a Managed Apple ID.
4. **Gate:** your two `PILOT – Shared iPad Users` must be **clear of conflict** before Step 12. If they are not, the pilot cannot run — pick two clear users or resolve theirs first.

Step 4. Federate ABM to Microsoft Entra ID

1. Sign in to `https://business.apple.com` as an ABM administrator.
2. Configure **federated authentication** against **Microsoft Entra ID** for your verified domain (e.g. `contoso.com`).
1. Complete the Microsoft-side consent when ABM redirects you — sign in with a **Global Administrator** and grant the requested permissions.
1. Confirm ABM shows federation as active for the domain.

> ***VERIFY WITH APPLE AND MICROSOFT.*** *The ABM navigation for federation, the flow name, the exact Microsoft permissions requested, how ABM handles a domain that still contains unresolved personal Apple IDs, and the behaviour for accounts that already have a Managed Apple ID are* ***Apple's and Microsoft's, and they change****. **Follow Apple's current ABM federation documentation for the exact steps.***
>
> ***Sequencing matters:*** *federate* ***after*** *Step 3 has cleared conflicts on the domain, or at minimum after you understand which addresses are conflicted. Federating a domain full of unresolved personal Apple*

IDs is how a project acquires a surprise.

***Related catalog solution:** if you want Managed Apple IDs created automatically from Entra ID rather than by hand, see *Implement Entra ID Integration with ABM for Account Auto Provisioning for Microsoft Intune*. It covers the provisioning of accounts into ABM; it does not cover Shared iPad.*

Phase 2 — Configure the Apple connection

Step 5. Create the Apple MDM Push (APNs) certificate

*(Skip if Intune already shows an **Active** certificate with a future expiry — do **not** create a second one, it will break your existing fleet.)*

1. Intune → **Devices** → **Enrollment** → **Apple** → **Apple MDM Push certificate**.
2. Check **I agree**, then **Download your CSR**.
3. Click **Create your MDM push certificate** → `https://identity.apple.com`.
4. Sign in with a **company-owned Apple ID**. **Record this Apple ID** — you must reuse the exact same one to renew every year. Never use a staff member's personal Apple ID.

1. **Create a Certificate**, upload the CSR, then **Download** the `.pem`.
2. Back in Intune, enter the **Apple ID** you used, **Browse** to the `.pem`, and click **Upload**.
3. Confirm the status reads **Active** with an expiry ~1 year out. Put the renewal in the operational calendar.

Step 6. Connect Apple Business Manager (upload the ADE token)

1. Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** → **Add**.
2. Check **I agree**, then **Download your public key** (the `.pem`).
3. Go to `https://business.apple.com` and create a new **MDM server** entry; **name it `Intune – Shared iPad`; upload the Intune public key; save it; then** download the token **(`.p7m`).** VERIFY WITH APPLE** — the ABM navigation to MDM server management changes; get the current path from Apple's ABM documentation.

1. Back in Intune, enter the **Apple ID** used for ABM, **Browse** to the `.p7m`, and click **Upload**.
2. Confirm the token appears under **Enrollment program tokens** with a **future** expiry.

This token expires annually — calendar the renewal now.

Step 7. Create the ADE enrollment profile with Shared iPad enabled

This is the step that delivers session isolation. Everything else is supporting work.

1. Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens**.
2. **Open the token** you uploaded in Step 6 (click its name — you must drill into the token; the Profiles tab does not exist anywhere else).

1. Select **Profiles** → **Create profile** → **iOS/iPadOS**.

2. **Name:** `SHARED IPAD – ADE Profile – Shared`. **Description:** `Shared iPad, supervised, field frontline fleet.`

1. **Platform:** confirm **iOS/iPadOS**. Click **Next**.

2. On the **Management settings** page:

• **User Affinity:** select `Enroll with user affinity`, then select `Shared iPad` as the enrollment mode. *(Shared iPad is presented under the user-affinity choice — it is a multi-user model, so it is not "without user affinity". If you cannot see the Shared iPad option, the token or the platform selection is wrong; see Troubleshooting §6.)*

• **Maximum number of cached users** (or the equivalent resident-user setting): set it to the number of shift users you need resident on the device. **VERIFY WITH APPLE** — the maximum Apple supports, and the storage each resident user consumes, are Apple's published figures and they change; confirm against Apple's current Shared iPad documentation and against Requirement R11 **before** you set this. Setting it above what the hardware can carry produces sign-in failures, not a warning.

• **Supervised:** `Yes`. *(Required — Requirement R9.)*

• **Locked enrollment:** `Yes`. *(Users cannot remove management. Essential for a field fleet.)*

1. Click **Next** to the **Setup Assistant customization** page. Hide every pane the field team does not need — **Restore**, **Apple ID** *(the Managed Apple ID sign-in is the Shared iPad sign-in screen, not this pane)*, **Siri**, **Privacy**, **Screen Time**, **App Analytics**, **Terms and conditions**, **Display Zoom** — so the iPad comes up straight at the Shared iPad sign-in screen.

1. Click **Next** → **Review + Create** → **Create**.

2. Confirm the profile appears in the token's **Profiles** list.

Step 8. Assign the iPads and the profile

1. In **ABM** → **Devices**, select the iPad **serial numbers** for the pilot.

2. Assign them to the MDM server `Intune – Shared iPad` (Step 6), and confirm. **VERIFY WITH APPLE** — the ABM device-assignment navigation changes; get the current path from Apple's docs.

1. In Intune, open the token → **Devices** → click **Sync**. **Apple rate-limits this sync** — allow several minutes and do not click it repeatedly.

1. When the serials appear, select them → **Assign profile** → choose **`SHARED IPAD – ADE Profile – Shared`** → confirm.

1. Verify each serial's **Profile status** shows the assignment.

> ***Devices that do not appear here are not in ABM*** *and cannot be ADE-enrolled — see Requirements R6 and Troubleshooting §6. This is where an existing retail-bought fleet reveals itself. Establish the real number now, before you commit to a rollout date.*

Phase 3 — Configure the device

Step 9. Create the field lockdown restrictions profile

1. Intune → **Devices** → **Configuration** → **Create** → **New policy**.

2. **Platform:** `iOS/iPadOS`. **Profile type:** `Templates` → `Device restrictions` → **Create**.

3. **Name:** `SHARED IPAD – Restrictions – Field Lockdown`. **Description:** `Field frontline Shared iPad lockdown.` Click **Next**.

1. On the **Configuration settings** page, set the following. *(Each of these applies **only** to a supervised device — Requirement R9. On an unsupervised iPad most of them are silently ignored, which is a common source of "restrictions not applying" — see Troubleshooting §6.)*

App Store, Doc Viewing, Gaming

- **App Store:** `Block` — staff cannot browse or install personal apps.
- **Installing apps:** `Block`.
- **Removing apps:** `Block` — staff cannot uninstall the CMMS app.
- **In-app purchases:** `Block`.
- **Explicit content in media:** `Block`.

Built-in apps

- **Safari:** `Block` *(unless the CMMS or schematics workflow needs the browser — confirm with the field team before blocking; a blocked Safari that the CMMS app relies on for SSO breaks the workload).*
- **Camera:** `Allow` — the field team photographs assets. This is part of the workload.
- **FaceTime:** `Block`.
- **Apple Music / Podcasts / News / Game Center:** `Block`.

Cloud and storage

- **Backup to iCloud:** `Block`.
- **iCloud Photo Library / iCloud Drive:** `Block` *(prevents field photos leaving to a personal library).*
- **Managed apps syncing to cloud:** `Block`.

Connected devices and sharing

- **AirDrop:** `Block` — a shared field device should not be a data-exfiltration path.
- **Bluetooth modification:** `Block` *(unless the field workload pairs a Bluetooth accessory — confirm first).*
- **Pairing with a host computer / Apple Configurator:** restrict per your management model.

Accounts and identity — this is the group that matters most

- **Account modification:** `Block` — **staff cannot add personal accounts to the device.** This is the setting that stops a shift worker signing their personal Apple ID or personal mail into the field iPad.

- **Modifying the passcode:** `Block`.
- **Modifying the device name:** `Block`.
- **Erase all content and settings:** `Block`.
- **Modifying restrictions:** `Block`.

Keyboard and dictation

- **Predictive keyboard / auto-fill:** `Block` — reduces cross-shift leakage of typed identity

and text into shared keyboard state.

1. Click **Next** through **Scope tags**.
2. On **Assignments**, click **Add groups** and select `SHARED IPAD – FMD – Pilot`. Click **Next**.

(Add `SHARED IPAD – FMD – Production` at Expand, Step 14 — not now.)

1. **Review + create** → **Create**.

> ***App allowlisting vs app blocking — which to use.*** *A **blocklist** ("hidden apps") names the apps to remove and lets everything else through — it fails the moment a new app exists. An **allowlist** ("allowed apps") names the apps permitted and hides everything else — it is the correct model for a locked-down field device and it is what this solution uses (Step 11).* ***The restrictions profile above blocks the *capabilities*; the allowlist controls the *app inventory*.*** *You need both: restrictions stop the App Store and account modification; the allowlist decides what is on the home screen. Do not try to do the job with only one of them.*

Step 10. Create the compliance policy

1. Intune → **Devices** → **Compliance** → **Create policy**.
2. **Platform:** `iOS/iPadOS` → **Create**.
3. **Name:** `SHARED IPAD – Compliance – iPadOS`. Click **Next**.
4. **Compliance settings:**

- **Device Health** → **Jailbroken devices:** `Block`.
- **Device Properties** → **Minimum OS version:** set your supported floor. **VERIFY WITH APPLE** — set

this to the current Shared iPad minimum from Apple's documentation (Requirement R10), not a remembered number.

- **System Security** → **Require a password to unlock mobile devices:** set per the customer's policy

for a shared field device. *(Under Shared iPad each user authenticates at sign-in; discuss the device-passcode requirement with the customer's security owner rather than assuming.)*

1. Click **Next**. On **Actions for noncompliance**, keep the default **Mark device noncompliant →

Immediately**.

1. **Assignments** → **Add groups** → `SHARED IPAD – FMD – Pilot` → **Next**.
2. **Review + create** → **Create**.

Step 11. Deploy the field app allowlist

`SHARED IPAD – Apps – Field Allowlist` is the app set below. Deploy each app as **Required** to `SHARED IPAD – FMD – Pilot`, then hide everything else.

11a. Add each app.

For each app in the field workload:

1. Intune → **Apps** → **All apps** → **Add**.
2. **App type:** `iOS/iPadOS store app` *(or `iOS/iPadOS volume purchased program app` if you

license it through ABM — VPP is preferred for a fleet, since it does not depend on a user's App Store identity, which a Shared iPad does not have).*

1. **Search the App Store** and select the app.

> ***Identify every app by its exact publisher string and bundle ID — never by name alone.*** *App Store search returns multiple near-identical results, and deploying the wrong one to 40 locked devices is a re-staging exercise. Record the exact publisher and bundle ID for each app in your build sheet before you search: | App | Publisher (exact string) | Bundle ID | Notes | |---|---|---|---| | Your CMMS app | *(from the vendor — confirm in writing)* | *(from the vendor)* | Confirm the vendor supports **Shared iPad**. Ask explicitly; do not assume. | | Schematics / document viewer | *(from the vendor)* | *(from the vendor)* | Must render your drawing formats offline. | | Microsoft Outlook | `Microsoft Corporation` | `com.microsoft.Office.Outlook` | Not "Outlook Lite", not a third-party mail client. | | Microsoft Teams | `Microsoft Corporation` | `com.microsoft.skype.teams` | Not "Teams for Life" / personal Teams. | | Microsoft Authenticator | `Microsoft Corporation` | `com.microsoft.azureauthenticator` | Required if you later layer Option B (SDM broker) — see the note below. | **Get the CMMS and viewer publisher/bundle ID from the vendor in writing.** Do not guess them from search results.*

1. Click **Select** → **Next** through the app information.
2. **Assignments** → **Required** → **Add group** → `SHARED IPAD – FMD – Pilot` → **Next**.
3. **Review + create** → **Create**.
4. Repeat for every app in the table.

11b. Restrict the home screen to the allowlist.

1. Intune → **Devices** → **Configuration** → **Create** → **New policy** → `iOS/iPadOS` → `Templates` → `Device restrictions` → **Create**.

1. **Name:** `SHARED IPAD – Apps – Field Allowlist`. Click **Next**.
2. Find the **Show or hide apps** settings group.
3. Select the **allowlist** control (`Allowed apps` / "Apps visible") — **not** the hidden-apps

blocklist.

1. Add **each** app by its **bundle ID** from the table in 11a, **plus** the system apps the workload

genuinely needs (Camera; Settings only if the field team must join Wi-Fi themselves — otherwise leave it out). Everything not listed is hidden.

1. **Assignments** → **Add groups** → `SHARED IPAD – FMD – Pilot` → **Next**.

2. **Review + create** → **Create**.

Do not lock the iPad to a single app. *Single-app (kiosk) mode is the right answer for a single-purpose appliance and the **wrong** answer here — this fleet runs a CMMS app **and** a viewer **and** the camera **and** Microsoft 365, and each shift user needs to reach the Shared iPad sign-out control. Use the **allowlist**, which is a lockdown that still permits the workload.*

If you are layering Option B (Microsoft 365 Shared Device Mode) on top of Shared iPad: *SDM requires the **Microsoft Authenticator** broker and per-app configuration, and it only affects apps that **participate** in SDM. **The set of Microsoft apps supporting shared device mode on iOS/iPadOS changes over time — verify the current supported-app list against Microsoft's documentation before designing around it.** This solution does not assert that list. Deploy and validate **Option A (Shared iPad) first**; it is what delivers the OS-level isolation the test in §5 measures. Layer SDM afterwards, and re-run the §5 test after you do.*

Phase 4 — Pilot

Step 12. Wipe and enroll the pilot iPad

Gate check before you touch a device: *the two members of* `PILOT – Shared iPad Users` *hold **Managed Apple IDs** and are **clear of Apple ID conflict** (Step 3). If not, stop — the pilot will fail at sign-in and you will spend a day debugging a policy that is fine.*

1. On the pilot iPad, back up anything the field team needs off it. **This step erases the device.**
2. **Settings** → **General** → **Transfer or Reset iPad** → **Erase All Content and Settings.** Confirm.
3. The iPad restarts into **Setup Assistant**. Choose language and region, and connect to the field

Wi-Fi (Requirement R17 — use a location with known-good signal).

1. Setup Assistant displays **Remote Management** naming your organization. Continue.
2. The iPad enrolls, supervises, and lands on the **Shared iPad sign-in screen** asking for a **Managed

Apple ID**.

• **If it lands on a normal personal-iPad home screen instead, it is not in Shared iPad mode.** Stop

and go to Troubleshooting §6 — this is the #1 failure and the entire solution depends on getting it right. Do not proceed and do not "fix it later".

1. In Intune → **Devices** → **All devices**, confirm the iPad appears; open it → **Properties** and

confirm **Supervised =** `Yes` and the enrollment profile is `SHARED IPAD – ADE Profile – Shared`.

1. Confirm the device is now a member of `SHARED IPAD – FMD – Pilot`. *(If the group is empty, the

Step 2b owner is missing — go to Troubleshooting §6.)*

1. Allow the device to sync. Confirm the restrictions, compliance, and the allowlist apps arrive.

Step 13. Run the validation gate

1. Run `SHAREDIPAD-Validate.ps1` and confirm an overall **PASS**.

2. **Run the session-isolation A/B test in Testing §5 with two real humans.** That test — not the

script — is the acceptance gate for this project. It is what the customer is buying.

1. **Do not proceed to Step 14 until the session-isolation test PASSES.** Expanding a broken isolation

model across 40 devices means wiping 40 devices twice.

Phase 5 — Expand

Step 14. Stage the fleet

1. **Confirm the eligible device count first.** In ABM → **Devices**, confirm **every** iPad serial in

the fleet is present. Serials that are absent are **not in ABM**, cannot be ADE-enrolled, and **cannot run Shared iPad** (Requirement R6). Establish this number before you schedule anything — it is the difference between a configuration project and a hardware purchase.

1. **Confirm storage eligibility.** Run `SHAREDIPAD-Report.ps1` and check the model and storage

columns against Apple's current guidance for your resident-user count (Requirement R11). Under-specced iPads must be re-sized or excluded — not hoped over.

1. **Assign the remaining eligible serials** to the `Intune – Shared iPad` MDM server in ABM, then in

Intune assign them the `SHARED IPAD – ADE Profile – Shared` profile (Step 8). *(Alternatively, set it as the token's **default** profile so newly added serials inherit it automatically.)*

1. **Extend the policy assignments to production.** For each of **`SHARED IPAD – Restrictions – Field

Lockdown, SHARED IPAD – Compliance – iPadOS, SHARED IPAD – Apps – Field Allowlist, and each app in the allowlist: open it → Properties → Assignments → Edit → Add groups → SHARED IPAD – FMD – Production → Review + save.

1. **Verify the Step 2b owner is on** `SHARED IPAD – FMD – Production` before the first wave enrolls.

Check it; do not assume it carried over.

1. **Stage in waves, not in one weekend.** Wipe and re-enroll in batches of 8–10 (Step 12) so a shift

is never without devices. After each wave, run `SHAREDIPAD-Validate.ps1` and spot-check the session-isolation test on one device from the wave **before** starting the next.

1. **Brief the field team** on the shift-handover runbook (Testing §5) before their devices are staged.

A user who does not know to sign out is the one operational gap Shared iPad still leaves you.

1. Run `SHAREDIPAD-Report.ps1` after the final wave and keep the CSV as the as-built record.

5. Testing & Validation — Shared iPad for Frontline Teams

This section is the point of the whole project. Everything in Implementation §3 exists so that **Test 5 — the session-isolation A/B test** passes. Run it with two real humans on a real pilot iPad before a single production device is staged.

Each test has an explicit **PASS / FAIL** gate. A FAIL is not a warning — it is a stop.

Test 1 — Tenant and Apple readiness

Procedure: Run `SHAREDIPAD-Discovery.ps1`.

PASS: The Apple MDM Push certificate is **Active** with a future expiry; an **ADE token** is present with a future expiry; an enrollment profile `SHARED IPAD – ADE Profile – Shared` exists and **Shared iPad is enabled** on it; `SHARED IPAD – FMD – Pilot` and `SHARED IPAD – FMD – Production` exist; `SHARED IPAD – Restrictions – Field Lockdown` and `SHARED IPAD – Compliance – iPadOS` exist; the allowlist apps are present.

FAIL: Any item missing → resolve it before piloting (Requirements §2, Implementation §3).

Test 2 — Enrollment-time grouping owner

Procedure: Entra → **Groups** → `SHARED IPAD – FMD – Pilot` → **Owners**. Repeat for `SHARED IPAD – FMD – Production`.

PASS: The Intune Provisioning Client service principal — matched by **AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`, not by display name — is listed as an **Owner** of **both** groups, and both groups are **Assigned** (not Dynamic).

FAIL: Missing owner, or a Dynamic group → Implementation §3 Step 2b. **The group will stay empty at enrollment and nothing will deploy.** Verify this before every wave.

Test 3 — The pilot iPad is actually in Shared iPad mode

This is the test that predicts Test 5. If this fails, Test 5 will fail, and you will waste an hour looking at Outlook.

Procedure:

1. On the enrolled pilot iPad, observe the screen it lands on after Setup Assistant.
2. In Intune → **Devices** → **All devices** → **(the pilot iPad)** → **Properties**.
3. Run `SHAREDIPAD-Validate.ps1`.

PASS: The device presents a **Shared iPad sign-in screen requesting a Managed Apple ID** — *not* a normal iPad home screen. In Intune, **Supervised =** `Yes`, the enrollment profile is `SHARED IPAD – ADE Profile – Shared`, and the device is enrolled via **ADE**. The device is a member of `SHARED IPAD – FMD – Pilot`.

FAIL: The device shows a normal home screen, or **Supervised =** `No`, or the device did not enroll via ADE → **the device is not in Shared iPad mode**. Go to Troubleshooting §6 — this is the #1 cause of the session leak, and no amount of app configuration will fix it. The device must be wiped and re-enrolled once the enrollment profile is correct.

Test 4 — Managed Apple ID sign-in works

Procedure: On the pilot iPad's Shared iPad sign-in screen, have **User A** (a member of `PILOT – Shared iPad Users`) sign in with their **Managed Apple ID**.

PASS: User A signs in. With federation (Requirement R13) they are prompted by the **Microsoft** sign-in page and use their normal work credentials. The device sets up their data space and lands on the home screen showing **only** the `SHARED IPAD – Apps – Field Allowlist` apps.

FAIL: Sign-in is rejected → the Managed Apple ID does not exist, or the address is still conflicted by a personal Apple ID (Requirement R14 / Implementation §3 Step 3). Sign-in hangs → check network (R16) and Wi-Fi at this location (R17). Sign-in reports the device is full → storage (R11). See Troubleshooting §6.

Test 5 — THE SESSION-ISOLATION A/B TEST *(the acceptance gate)*

This is the test the customer is buying. It is the reason the project exists. Run it with **two real people**, on a **real pilot iPad**, in the **field Wi-Fi environment**. Do not simulate it, do not run it with one person signing in twice, and do not sign the assessment off without it.

You need: the pilot iPad (Test 3 PASSED), **User A** and **User B** — two different members of `PILOT – Shared iPad Users`, both with working mailboxes (Requirement R2) and Managed Apple IDs (Test 4 PASSED).

Part 1 — User A creates a session with real data

1. **User A** signs in to the iPad with their **Managed Apple ID**.
2. Open **Microsoft Outlook**. Sign in if prompted. Wait for the mailbox to sync.
3. **Send an email** from User A to themselves with the subject `ISOLATION TEST A`. Confirm it arrives in their inbox. *(Now there is unmistakable, identifiable data on the device.)*
1. Open **Microsoft Teams**. Sign in if prompted. **Send a chat message** to a colleague containing the text `ISOLATION TEST A`. Confirm it appears in the chat list.
1. Open the **Camera** and **take a photo** of something identifiable. Confirm it appears in Photos.
2. Open the **CMMS app**. Sign in and open any work order. *(This is your real workload — test it.)*
3. Open the **schematics / document viewer** and open a drawing.
4. **User A signs out of the Shared iPad session** — use the Shared iPad sign-out control (the user picker / Sign Out), **not** the app-level sign-out inside Outlook. *(App-level sign-out is exactly the thing that does not work. Test the real mechanism.)*
1. Confirm the device returns to the **Shared iPad sign-in screen**.

Part 2 — User B signs in on the SAME device

User B now signs in on the same physical iPad and must observe *nothing* of User A's.

Check each of the following. **Every single one must pass.**

#	Check	PASS	FAIL
5.1	**Outlook mailbox** — open Outlook	Outlook prompts for sign-in, **or** opens **User B's** mailbox. There is **no** mailbox belonging to User A.	User A's mailbox is present.
5.2	**Cached mail** — search for `ISOLATION TEST A` in Outlook	**Zero results.** The message does not exist on this device for User B.	The message is visible or cached.
5.3	**Teams chat** — open Teams	Teams prompts for sign-in, **or** opens **User B's** chats. `ISOLATION TEST A` is **not** present.	User A's chat history is visible.
5.4	**Photos** — open Photos / Camera roll	**Empty**, or contains only User B's own photos. User A's photo is **not** present.	User A's photo is visible.
5.5	**Identity autofill** — open Outlook / Teams / the CMMS sign-in and tap the username field	No suggestion of **User A's** address. The field is empty or offers only User B's.	User A's address is offered by autofill or is pre-filled.
5.6	**Signed-in state** — check every app in the allowlist, including the **CMMS app** and the **viewer**	**Nothing** is silently signed in as User A. Each app is signed out or signed in as User B.	Any app opens already signed in as User A.
5.7	**Files / documents** — open the viewer's recent documents and the Files app	No document User A opened is present or accessible to User B.	User A's documents are listed or openable.
5.8	**Safari** *(only if Safari is allowed in your build)* — open it and check history and autofill	No history or saved identity from User A's session.	User A's browsing state persists.

The verdict

• **PASS = every one of 5.1 through 5.8 passes.** Not most. Every one. A single leak means the architecture is not delivering isolation, and one leaked mailbox is the whole problem you were hired to fix.

• **Now reverse it.** Sign User B out, sign **User A** back in, and confirm A gets **their own** data back (their mail, their photo) and **none** of B's. Isolation must work in both directions — a device that gives everyone a blank slate but loses each user's own data is also broken.

What a FAIL means — read this before you debug

A FAIL is almost never an app problem. Diagnose in this order:

Symptom	Which architecture step is wrong	Fix
Everything leaks — mail, photos, chat, all of it	**The device is not in Shared iPad mode.** It is a single-user iPad being handed between people — the original problem, unchanged. This is the #1 cause.	Re-run **Test 3**. Check the device is **ADE-enrolled**, **Supervised = Yes**, and the enrollment profile is `SHARED IPAD – ADE Profile – Shared` **with Shared iPad enabled** (Implementation §3 Step 7). Wipe and re-enroll (Step 12).
The device never asked for a Managed Apple ID — it went to a home screen	Same as above — Shared iPad was never enabled, or the device did not enroll via ADE.	Implementation §3 Steps 7–8, then Step 12. Confirm the serial is in ABM (R6).
Photos leak but mail is clean	The device **is** in Shared iPad mode, but something outside the per-user data space is being written — or you are looking at an app that stores outside it.	Confirm iCloud Photo Library is **Blocked** (Step 9) and re-test. Escalate to the app vendor if it persists.
Only the CMMS app leaks — Microsoft apps are clean	**The CMMS app is not honouring the Shared iPad user boundary.** Shared iPad is working; that one app is not designed for it.	Take it to the vendor with this test as evidence. Ask explicitly whether the app supports **Apple Shared iPad**. This is a vendor gap, not a configuration gap — and you cannot configure around it.
Only Microsoft apps leak — photos and files are clean	Shared iPad is working at the OS layer. This is unexpected and is worth a support case.	Re-verify Test 3, then raise it with Microsoft. Do **not** reach for Shared Device Mode as a patch until you understand why — layering B on a broken A hides the real fault.
User B sees a sign-in prompt but User A's address is pre-filled	Keyboard/autofill state, not session state. Minor but real — it is a data disclosure.	Confirm **predictive keyboard / auto-fill** is **Blocked** in `SHARED IPAD – Restrictions – Field Lockdown` (Step 9) and re-test.

Test 6 — Field lockdown is enforced

Procedure: As User B, on the pilot iPad, attempt each of the following.

PASS: The **App Store** is not present and cannot be reached. Apps **cannot** be installed or removed. **Adding an account** (personal Apple ID, personal mail) is **blocked**. **Erase All Content and Settings** is unavailable. **AirDrop** is unavailable. Only the `SHARED IPAD – Apps – Field Allowlist` apps appear on the home screen. **Camera works** (the field team needs it).

FAIL: Any restriction not enforced → confirm **Supervised =** `Yes` (Test 3 — unsupervised devices silently ignore most of these), confirm the profile assignment to `SHARED IPAD – FMD – Pilot`, and force a

sync. See Troubleshooting §6.

Test 7 — Compliance reports correctly

Procedure: Intune → **Devices** → **Compliance** → `SHARED IPAD – Compliance – iPadOS` → **Device status**.

PASS: The pilot iPad reports **Compliant** against the policy.

FAIL: Not evaluated → the device is not in `SHARED IPAD – FMD – Pilot` (Test 2). Non-compliant → read the reason; a minimum-OS failure means the device needs an iPadOS update (R10).

Test 8 — The field workload actually works

Procedure: As User B, complete a genuine shift task end-to-end: open a work order in the **CMMS app**, open a **schematic** in the viewer, **take a photo** and attach it to the work order, and send a **Teams** message about it.

PASS: The complete workload succeeds on a locked-down Shared iPad.

FAIL: A step is blocked → a restriction in `SHARED IPAD – Restrictions – Field Lockdown` is too tight (Safari and Bluetooth are the usual culprits), or an app is missing from `SHARED IPAD – Apps – Field Allowlist`. Fix and re-test — **and then re-run Test 5**, because loosening a restriction can reopen a leak.

Test 9 — Storage holds under multiple resident users

Procedure: Sign three or four different pilot users into the same iPad in sequence, each doing a short real task. Then check Intune → **Devices** → **(device)** → **Hardware** for free storage, and run `SHAREDIPAD-Report.ps1`.

PASS: All users sign in successfully and free storage remains healthy against Apple's current guidance for your resident-user count.

FAIL: A sign-in fails for lack of space, or users are being evicted → the hardware is undersized (Requirement R11) or the cached-user setting exceeds what the device can carry (Implementation §3 Step 7). **VERIFY WITH APPLE** — check the current storage and maximum-user guidance in Apple's Shared iPad documentation. **Find this out at the pilot, not at device 30 of 40.**

Acceptance gate

Do not stage a single production device until:

- Tests **1, 2, 3, 4** PASS *(the architecture is real)*
- Test **5** PASSES **completely — all of 5.1 to 5.8, in both directions** *(the leak is fixed — this is the gate)*
- Tests **6, 7, 8, 9** PASS *(the device is locked down, healthy, usable, and sized)*

If Test 5 fails, **nothing else matters**. Fix the architecture and re-run it.

Shift-handover runbook — for the field team

Print this. Put it on the iPad rack. Brief the team before their devices are staged.

Shared iPad gives you a clean device — but only after you sign out. A user who walks away without signing out leaves their session open on the rack for the next person to walk into.

At the END of your shift

1. **Close what you are working on.** Save and submit any open work order in the CMMS app.
2. **Sign out of the iPad session.** Tap your name / user picker → **Sign Out**.

• **This is not the same as signing out of Outlook.** Signing out of an app does not end your session. Use the **iPad's** sign-out.

1. **Wait for the sign-in screen to appear** before you put the iPad down. Sign-out has work to do — walking away mid-way can leave your session open.

1. **Return the iPad to the rack and put it on charge.** A dead iPad is an iPad the next shift cannot use.

At the START of your shift

1. **Take any iPad from the rack.** They are interchangeable — you will get your own environment.
2. **Check the sign-in screen is showing.** If you see **someone else's** home screen or name, the previous user did not sign out — sign them out via the user picker, or report it. **Do not use the device inside their session.**

1. **Sign in with your work account** (your Managed Apple ID). You will get your own mail, your own photos, and your own work orders.

1. **Stand where the Wi-Fi is good** when you sign in. Sign-in needs the network.

Report it if

• You sign in and see **anyone else's** mail, messages, photos, or work orders. **Report it immediately** — that is exactly what this system exists to prevent, and it means something is broken.

• The iPad says it is full, or will not let you sign in.

• Your own mail or photos are missing after you sign in.

6. Troubleshooting — Shared iPad for Frontline Teams

Symptom → cause → fix. Work top-down: the first entry is the cause of the overwhelming majority of failures in this solution, and almost every other symptom in this section is a downstream consequence of it.

1. "The next user still sees the prior user's mail" — THE #1 CAUSE

Symptom: User B signs in (or picks up the iPad) and sees User A's mailbox, Teams chat, photos, or documents. Test 5 fails. This is the original problem, unchanged.

Cause — in order of likelihood:

*1a. The device is not actually in Shared iPad mode *(the cause ~90% of the time)**

The iPad is a **single-user device being handed between people**. The OS has no per-user boundary, so there is nothing to isolate. Everything you configured downstream is irrelevant — restrictions, compliance, and app settings cannot create a session boundary that the OS does not have.

Diagnose:

1. **Look at the device.** After enrollment, does it present a **Shared iPad sign-in screen asking for a Managed Apple ID? If it lands on a** normal iPad home screen**, it is** not** in Shared iPad mode. Stop here — this is your answer.

1. Intune → **Devices** → **All devices** → **(the iPad)** → **Properties**. Check:

• **Supervised =** `Yes`? If `No`, the device did not go through ADE.

• **Enrollment profile =** `SHARED IPAD – ADE Profile – Shared`?

• **Enrollment type** — is it an Apple/ADE (DEP) enrollment type, or a manual/user enrollment?

1. Run `SHAREDIPAD-Validate.ps1`. Look specifically at **"Shared iPad ENABLED on that profile"**.

2. Run `SHAREDIPAD-Report.ps1` and check the device's **ProfileIsShared** and **SharedIPadReady** columns.

Fix — depending on what you found:

What you found	What it means	Fix
The enrollment profile does **not** have Shared iPad enabled	The profile is wrong. Every device enrolled with it is single-user.	Fix the profile (Implementation §3 Step 7), **then wipe and re-enroll every affected device** (Step 12). Changing the profile does **not** convert already-enrolled devices.
The profile is correct but the device didn't get it	The serial was never assigned the profile.	Assign it on the token (Step 8), then **wipe and re-enroll** the device (Step 12).
Supervised = `No` / the device enrolled manually	The device never went through ADE.	**Wipe and re-enroll via ADE** (Step 12). There is no in-place conversion — you cannot make an enrolled iPad supervised or Shared without erasing it.
The serial isn't on the ADE token at all	The device is not in ABM.	See §6 below. This may be a procurement problem.

> ***The critical fact: you cannot convert a device into Shared iPad mode in place.*** *Shared iPad is established **at enrollment**. Fixing the profile and waiting for a policy sync does nothing. The device*

*must be **erased and re-enrolled** through ADE. If you take one thing from this section, take this.*

*1b. The device *is* in Shared iPad mode, but users aren't signing out*

Diagnose: Watch a real handover. Does the outgoing user use the **iPad's** sign-out (the user picker → Sign Out), or do they just put the iPad down? Does the device show the sign-in screen when it's on the rack, or someone's home screen?

Fix: Brief the field team on the shift-handover runbook (Testing §5) and put it on the rack. Shared iPad gives a clean device **after** sign-out. A user who walks away leaves an open session — the one gap the architecture does not close for you. Also confirm the handover point has good Wi-Fi (R17): a sign-out attempted on a dead spot may not complete.

1c. It's one specific app leaking, not the device

Diagnose: Run the Test 5 checklist item by item. If Outlook, Teams, and Photos are clean and only the **CMMS app** (or the viewer) leaks, Shared iPad is working — that app is not honouring the per-user boundary.

Fix: This is a **vendor gap and you cannot configure around it**. Take the Test 5 results to the app vendor as evidence and ask explicitly: *does this app support Apple Shared iPad?* Do not spend days in Intune on it — there is no setting.

2. Managed Apple ID sign-in fails

Symptom: A user cannot sign in at the Shared iPad sign-in screen.

Symptom detail	Cause	Fix
The account is not recognised / rejected	**The user has no Managed Apple ID.** Usually because their address is still conflicted by a personal Apple ID (R14).	Confirm in ABM → **Users** that the Managed Apple ID exists. If not, resolve the conflict (§3 below / Implementation Step 3). **This is a gate — pick a clear user for the pilot rather than fighting it mid-test.**
Sign-in hangs or times out	Network. Apple's endpoints are unreachable, or the Wi-Fi at this location is poor (R16/R17).	Test from the same spot: reach `https://login.microsoftonline.com`. Confirm Apple's published enterprise endpoints are allowed through the proxy. **VERIFY WITH APPLE** — use Apple's current published network requirements list, not an older allowlist. Retry at a known-good location to isolate it.
"Device is full" / sign-in refused for space	Storage (R11). See §7 below.	§7.
The wrong sign-in page appears, or none	Federation is not configured or not active for the domain.	§4 below.

Symptom detail	Cause	Fix
Password rejected but the account exists	If **not** federated, the Managed Apple ID has its own separate password — the user is typing their work password.	Either federate (R13, Implementation Step 4) or issue the Managed Apple ID password. **Federate.** A second password on a shared frontline device ends up written on the rack.

3. ABM says "address already in use" / the Managed Apple ID cannot be created

Symptom: ABM will not issue a Managed Apple ID at a staff member's work address.

Cause: A personal Apple ID is already using that address. ABM cannot issue a Managed Apple ID at an address in use by a personal Apple ID. This is expected, it is Requirement R14, and it is the single largest blocker in this project.

Fix:

1. Run `SHAREDIPAD-AppleIDConflict.ps1` to build the **candidate** worklist from Entra ID.

Remember what this list is and is not: it tells you who *would need* a Managed Apple ID and at which address. **It cannot query Apple and cannot tell you which addresses actually conflict.**

1. **Get the authoritative conflict list from Apple Business Manager.** ABM is the only source that knows. Reconcile it into the CSV's `ABMConflict` column.

1. Resolve the conflicts via Apple's supported path, which is built around **verifying your domain** in ABM; affected personal Apple IDs must then be changed to a personal email address (or are otherwise handled by Apple's process).

> ***VERIFY WITH APPLE.*** *The exact flow name, where it lives in ABM, what ABM reports, the notification Apple sends the affected user, the* ***grace period****, and* ***what happens to a user who does not act*** *are* ***all Apple's and all change****.* ***Get every one of these from current Apple Business Manager documentation before you plan a date or write user communications.*** *A wrong timeline here is a broken promise to staff.*
>
> ***Before you act on any of this:*** *whether to require staff to rename or surrender a personal Apple ID tied to their work address is an* ***identity, HR, legal, and communications decision*** *— it reaches their personal iPhone, personal Mac, iCloud photo library, and App Store purchases.* ***Present the options and their impact to the customer's HR, legal, and identity owners and let them decide*** *(Implementation Step 3c). Do not make the call. Do not send a staff notification without executive sponsorship and an agreed comms plan and grace period (R18).*

Tactical workarounds while the campaign runs:

• **For the pilot:** pick two `PILOT – Shared iPad Users` who are **clear of conflict**. Do not burn the pilot on a blocked user — you will end up debugging a configuration that is fine.

• **Worth raising early:** issuing Managed Apple IDs on a **verified subdomain** the staff have never used personally (e.g. `staff.contoso.com`) may sidestep many conflicts entirely. **VERIFY WITH APPLE** that this fits your ABM and federation configuration before proposing it — but raise it *before* the

campaign starts, because switching to it later is expensive.

4. Federation sign-in loop, or the Microsoft sign-in page never appears

Symptom: Signing in with a Managed Apple ID loops between Apple and Microsoft, returns to the sign-in screen repeatedly, or never presents the Microsoft page at all.

Cause / fix:

Cause	Fix
Federation is not active for that domain	In ABM, confirm federated authentication is configured **and active** for the exact domain of the address being used (e.g. `contoso.com`).
The address is on a domain you never federated	Users on a second domain or a subdomain need that domain verified and federated too. Check the exact address the user is typing.
The Entra account is disabled, blocked, or has no licence	Entra → **Users** → **(the user)**: account enabled, and licensed (R1/R2). A disabled account cannot complete federated sign-in.
Conditional Access is blocking the sign-in	Entra → **Sign-in logs**, filter to the user, and read the **Conditional Access** tab. A CA policy requiring a compliant device can block the sign-in that is *establishing* the session on that device. Move the offending policy to **report-only** to confirm, then fix the policy — do not leave it off.
The Microsoft consent was never completed	Re-run the ABM federation setup and complete the Microsoft-side consent as a **Global Administrator** (Implementation Step 4).
The user still has an unresolved personal Apple ID at that address	Federation does not resolve conflicts. §3 above.

> ***VERIFY WITH APPLE AND MICROSOFT.*** *The ABM federation flow, its navigation, the exact Microsoft permissions requested, how ABM behaves for a domain with unresolved personal Apple IDs, and the handling of pre-existing accounts are* ***Apple's and Microsoft's and they change****. Follow Apple's* ***current*** *ABM federation documentation rather than a remembered flow.*

5. Restrictions not applying / the App Store is still there

Symptom: `SHARED IPAD – Restrictions – Field Lockdown` shows as assigned, but the device still has the App Store, still allows adding accounts, or still allows app removal.

Cause	Diagnose	Fix
The device is not supervised *(the usual cause)*	Intune → **Devices** → **(device)** → **Properties** → **Supervised**.	Most of these restrictions **only apply to supervised devices** and are **silently ignored** on an unsupervised one. **Wipe and re-enroll via ADE** with Supervised enabled (Steps 7, 12). No in-place fix exists.
The device isn't in the group	Intune → **Devices** → **(device)** → **Group membership**; or run `SHAREDIPAD-Validate.ps1`.	If the group is **empty**, go to §8 — the Provisioning Client owner is missing.
The profile isn't assigned to *this* group	`SHAREDIPAD-Validate.ps1` reports assignment per group. Common at Expand: assigned to Pilot, forgotten for Production.	Add the assignment (Step 14), or run `SHAREDIPAD-Set-Restrictions.ps1 -DeviceGroupName 'SHARED IPAD – FMD – Production' -WhatIf:$false`.
Not synced yet	Intune → **Devices** → **(device)** → **Sync**; check the profile's **Device status**.	Allow time and re-check. Policy delivery is not instant.
Profile shows **Error** or **Conflict**	Open the profile → **Device status** → the device → read the per-setting result.	A **Conflict** means another profile sets the same setting differently. Find the other profile and resolve it — the device does not pick a winner for you.
Camera/Safari blocked and the workload broke	The lockdown is too tight.	Loosen the specific setting (Step 9), **then re-run Test 5** — loosening a restriction can reopen a leak.

6. The device won't ADE-enroll / the serial isn't on the token

Symptom: Setup Assistant never shows **Remote Management**; the iPad comes up as a normal personal device; or the serial does not appear under the ADE token in Intune.

Cause	Diagnose	Fix
The device is not in Apple Business Manager *(the most common, and the most expensive)*	ABM → **Devices** → search the serial. No result = not in ABM.	**A device not in ABM cannot be ADE-enrolled and therefore CANNOT run Shared iPad** (R6). There is no Intune setting that fixes this. Either bring it into ABM by an Apple-supported route, or replace it with an iPad purchased through an ABM-linked reseller. **VERIFY WITH APPLE** — the routes for adding already-owned devices, the eligibility rules, and any provisional window are Apple's and change. **Establish this across the whole fleet in the Assess phase** — it decides whether this is a configuration project or a hardware purchase.
The serial isn't assigned to the MDM server	ABM → **Devices** → check the serial's assigned MDM server.	Assign it to `Intune – Shared iPad`, then **Sync** in Intune (Step 8).
Assigned in ABM but not visible in Intune	Intune → token → **Devices** → **Sync**.	Apple **rate-limits** this sync. Wait several minutes. **Do not click Sync repeatedly** — it will not go faster.
The serial has no enrollment profile assigned	Intune → token → **Devices** → check the **Profile status** column.	Select the serials → **Assign profile** → `SHARED IPAD – ADE Profile – Shared` (Step 8).
The device was already set up	It went past Setup Assistant.	**Erase All Content and Settings** (Step 12) so it returns to Setup Assistant.
The ADE token has expired	`SHAREDIPAD-Discovery.ps1` reports the expiry.	**The token expires annually.** Renew it in ABM and re-upload (Step 6). An expired token stops all new enrollments. Calendar it.
The APNs certificate has expired	`SHAREDIPAD-Discovery.ps1` reports the expiry.	Renew it **with the same company Apple ID** you originally used (Step 5). A new Apple ID means re-enrolling every managed device in the tenant.
Enrollment restrictions block iOS/iPadOS	Intune → **Devices** → **Enrollment** → **Enrollment device platform restrictions**.	Allow iOS/iPadOS for the target scope.

The Shared iPad option is missing from the enrollment profile: confirm you are creating an **iOS/iPadOS** profile, and that you drilled into the **ADE token** first (**Devices** → **Enrollment** → **Apple** →

Enrollment program tokens → **(open the token)** → **Profiles** → **Create profile** → **iOS/iPadOS**). The Profiles tab does not exist anywhere else in the console, and Shared iPad appears under the **user-affinity** choice on the Management settings page — it is a multi-user model, so it is not under "enroll without user affinity". **VERIFY WITH APPLE AND MICROSOFT** — the console layout for this page changes.

7. Storage full / a user can't sign in for lack of space

Symptom: A user cannot sign in and the device reports it is full; or users' data is being evicted and they have to re-download everything each shift; or later users in a shift cycle fail while early ones succeed.

Cause: Each resident user consumes storage on the device. The device is either undersized, or the enrollment profile's **maximum cached users** is set higher than the hardware can carry. This is the constraint teams underestimate, and it typically appears at wave 3, not at the pilot.

Diagnose:

1. Intune → **Devices** → **(device)** → **Hardware** — total and free storage.
2. Run `SHAREDIPAD-Report.ps1` and review the `StorageGB` / `FreeGB` / `FreePercent` columns across the

fleet. Look for a pattern, not a single device.

1. Check the **maximum cached users** on `SHARED IPAD – ADE Profile – Shared` (Step 7).

Fix:

• **Reduce the maximum cached users** to what the hardware can genuinely carry, and re-enroll the

affected devices for the new value to take effect at enrollment.

• **Or re-size the hardware.** Under-specced iPads are a procurement problem, not a configuration one.

• **VERIFY WITH APPLE** — Apple publishes the **current minimum and recommended storage** for Shared

iPad and the **maximum number of resident users**, and both have changed across iPadOS releases. **Get the figures from Apple's current Shared iPad documentation.** Do not use a number from an older deployment; do not use a number from this document — it deliberately does not give you one.

• **Run Test 9 at the pilot**, with three or four users on one device. Finding this at device 4 of 40 is

a conversation. Finding it at device 34 is a re-procurement.

8. The device group is empty after enrollment / nothing deploys

Symptom: iPads enroll successfully, but `SHARED IPAD – FMD – Pilot` (or `– Production`) is empty. No restrictions, no compliance, no apps arrive. Everything in the console looks correctly configured.

Cause: The Intune Provisioning Client service principal is not an Owner of the group, or the group is **Dynamic** instead of Assigned.

Enrollment-time device grouping works by that first-party principal **owning** the Assigned group. Without the owner, nothing is ever added to the group — and because every policy is assigned to the group, nothing deploys. **This failure looks exactly like a policy problem and is not one.**

Diagnose:

1. Entra → **Groups** → `SHARED IPAD – FMD – Pilot` → **Owners**.
2. Run `SHAREDIPAD-Validate.ps1` — it checks this explicitly and matches by AppId.

Fix:

1. Confirm the group's membership type is **Assigned**, **not Dynamic**. If it is Dynamic, this

mechanism does not apply at all — recreate it as Assigned (Step 2).

1. Entra → **Groups** → **(the group)** → **Owners** → **Add owners**, and search by **AppId

`f1346770-5b25-470b-88bd-d5744ab7952c`**.

• **Search by the AppId, not the display name.** The display name varies by tenant — many show

`Intune Autopilot ConfidentialClient`, others show something else. Matching on the display name is how people conclude the principal "isn't there" when it is.

• **If the principal genuinely isn't present:** open `https://intune.microsoft.com` once as a

Global Administrator to provision it into the tenant, then add it as owner.

1. **Verify the owner on** `SHARED IPAD – FMD – Production` **too**, before the first expansion wave. Do not

assume it carried over from the pilot group — it does not.

1. Re-enroll a test device to confirm it now lands in the group.

Verify this before every pilot and every wave. It is one click to check and a lost afternoon to diagnose.

9. An app isn't available or misbehaves under Shared iPad

Symptom: An app in `SHARED IPAD – Apps – Field Allowlist` doesn't install, doesn't appear, or behaves oddly — won't stay signed in, loses data between sessions, or leaks between users.

Symptom	Cause	Fix
App never installs	Not assigned **Required** to the device's group; or not synced.	Confirm the Required assignment to the correct group (Step 11a) and sync. `SHAREDIPAD-Validate.ps1` checks this.
App installs but isn't on the home screen	It's not in the **allowlist** (`SHARED IPAD – Apps – Field Allowlist`, Step 11b) — the allowlist hides everything it doesn't name.	Add the app's **bundle ID** to the allowlist.

Symptom	Cause	Fix
The wrong app installed	Selected by name from App Store search. Search returns near-identical results.	Remove it and re-add by the **exact publisher string and bundle ID** (Step 11a). Get the CMMS/viewer publisher and bundle ID **from the vendor, in writing**.
App won't install on a Shared iPad specifically	A store app may depend on an App Store identity that a Shared iPad does not have.	Re-add it as a **volume purchased (VPP/ABM)** app and license it **by device**. This is the preferred model for a fleet.
App loses the user's data between their own sessions	The app isn't storing data in the per-user space.	Vendor issue. Raise it with the vendor.
The app leaks data between users	**The app is not honouring the Shared iPad user boundary.**	**Vendor gap — you cannot configure around it.** Take the Test 5 evidence to the vendor and ask explicitly whether the app supports **Apple Shared iPad**. Do not burn days in Intune.
A Microsoft app behaves unexpectedly	Possible, and unexpected if Shared iPad is working.	Re-verify Test 3 first. Then raise a Microsoft support case. **Do not reach for Shared Device Mode as a patch** until you understand why — layering B on a broken A hides the real fault.

> ***If you are layering Option B (Microsoft 365 Shared Device Mode):*** *it requires the* ***Microsoft Authenticator*** *broker and per-app configuration, and it affects* ***only apps that participate in SDM****.* ***The set of Microsoft apps supporting shared device mode on iOS/iPadOS changes over time — verify the current supported-app list against Microsoft's documentation before designing around it.*** *This solution does not assert that list, and neither should you. An app that does not participate is simply unaffected by SDM — that is not a bug, and no amount of configuration will change it.*

10. Compliance shows Not evaluated / the device is Not compliant

Symptom	Cause	Fix
Not evaluated	The device isn't in the group the policy is assigned to.	§8, then confirm the assignment (Step 10).

Symptom	Cause	Fix
Not compliant — minimum OS	iPadOS is below the floor set in `SHARED IPAD – Compliance – iPadOS`.	Update iPadOS (R10). If the floor itself is wrong, **VERIFY WITH APPLE** the current Shared iPad minimum and correct the policy — don't lower the floor to make a red dashboard green.
Not compliant — jailbreak	Detected or falsely detected.	Investigate. On a supervised, ADE-enrolled, locked-down field iPad, a genuine jailbreak is a serious finding — treat it as one.
Not compliant — passcode	The policy requires a passcode the shared-use model doesn't set.	Discuss with the customer's security owner. Under Shared iPad each user authenticates at sign-in; decide the device-passcode requirement deliberately (Step 10) rather than by default.
Compliance grace period masking failures	The policy's noncompliance action is delayed.	Set **Mark device noncompliant** → **Immediately** for a field fleet so you see reality.

11. The scripts

Symptom	Cause	Fix
`Connect-MgGraph` fails or scopes are refused	The signed-in account lacks the rights, or admin consent hasn't been granted for the scope.	Sign in as **Intune Administrator** (or Global Administrator). Each script's header lists its exact **Required Graph scopes** — grant those, no more.
"Group not found" / **"profile not found"** for an object you can see in the portal	**The scripts match on EXACT display name, and the object names use an en dash (–), not a hyphen (-).** They look nearly identical and are different characters.	Copy the name **verbatim** from this guide, or pass it with the `-DeviceGroupName` / `-ProfileName` parameter. Do not retype the dash. This is the single most common script failure.
"Found N objects named X"	A duplicate object exists.	Resolve the duplicate in the portal. The scripts deliberately refuse to guess which one you meant.

Symptom	Cause	Fix
`SHAREDIPAD-Set-Restrictions.ps1` reports what it would do but changes nothing	**Working as designed — it defaults to `-WhatIf`.**	Read the dry-run output, then re-run with `-WhatIf:$false` to apply. That explicit step is the point.
`SHAREDIPAD-Validate.ps1` says PASS but Test 5 still fails	**Expected and important.** A PASS proves the **architecture** is configured. It cannot prove sessions are isolated.	Go to §1. The session-isolation A/B test (Testing §5) is the only real gate. Never sign a project off on the script alone.
`SHAREDIPAD-AppleIDConflict.ps1` shows every user as "UNKNOWN" conflict	**Working as designed. The script cannot query Apple.**	It builds the **candidate** list from Entra. **The authoritative conflict list comes from Apple Business Manager.** Fill the `ABMConflict` column from ABM. Never present the script's output to a customer as a conflict list.
"Specify a scope" error from `SHAREDIPAD-AppleIDConflict.ps1`	Deliberate — it refuses to run unscoped across a whole tenant.	Pass `-GroupName 'PILOT – Shared iPad Users'` or `-Domain 'contoso.com'`. An unscoped worklist is one nobody can action.
Shared iPad state reports "unknown" / a property looks absent	The Graph property naming for Shared iPad on the ADE enrollment profile has varied across API versions. The scripts check the known variants.	**Verify the true state in the portal**: Intune → **Devices** → **Enrollment** → **Apple** → **Enrollment program tokens** → **(token)** → **Profiles** → `SHARED IPAD – ADE Profile – Shared`. The portal is authoritative; treat a script "unknown" as *check by hand*, not as a failure.
Device counts don't match the portal	Graph paging, or a stale sync.	The scripts follow `@odata.nextLink` and page fully. If the numbers still differ, force a device sync and re-run.
Nothing returned for iPads	The filter looks for `operatingSystem eq 'iOS'` and a model containing "iPad" — correct for iPadOS, which Graph still reports as iOS.	If genuinely empty, no iPads are enrolled yet. Check Intune → **Devices** → **All devices**.

7. Best Practices — Shared iPad for Frontline Teams

Architecture

- **Fix the architecture, not the symptom.** Every hour spent tuning Outlook settings on a single-user

iPad passed between shifts is an hour wasted. There is no app configuration that produces per-user data separation on a device that has no per-user boundary. Deploy Shared iPad, or accept a partial fix and say so.

- **Decide A vs B explicitly, in writing, with the customer.** Shared iPad (OS-level) and Microsoft 365

Shared Device Mode (app-level) solve different amounts of the problem. Write the decision and its rationale down. A customer who thinks they bought OS-level isolation and got app-level isolation has a grievance you will be defending in six months.

- **Deploy Option A first, and validate it, before layering Option B.** Layering SDM on a broken Shared

iPad deployment hides the real fault and makes the failure harder to diagnose. Get the session test passing on Shared iPad alone, then add SDM and re-run the test.

- **Never assert the Microsoft SDM supported-app list from memory.** It changes. Check Microsoft's

current documentation each time you design around it. A design built on a stale list fails at validation, in front of the customer.

- **Never carry Apple's numbers forward from an old project.** The Shared iPad minimum iPadOS version,

the storage minimums, and the maximum resident-user count change. Check Apple's current documentation every time. This is the single most common way an experienced engineer gets it wrong.

The Apple ID reclamation campaign

- **Run the conflict discovery FIRST, before you promise a date.** The size of the conflict list

determines the project timeline more than any technical task. Find out on day one, not week six.

- **Never let the technical team make the reclamation decision by default.** Asking staff to change a

personal Apple ID reaches their personal iPhone, their family photos, and their App Store purchases. That is an HR, legal, and identity decision. Put the options in front of the customer's HR, legal, and identity owners, get the decision in writing, and then execute it. "The engineer ran a script and everyone got an email from Apple" is how a project becomes an incident.

- **Get executive sponsorship before the first notification goes out.** No sponsor, no comms plan, no

agreed grace period → do not send anything. This is Requirement R18 and it is the one most often skipped.

- **Consider the subdomain option early.** Issuing Managed Apple IDs on a subdomain staff have never

used personally (e.g. `staff.contoso.com`) may sidestep the conflict entirely and turn a personal-data confrontation into a non-event. **Verify with Apple** that it fits your ABM/federation design before proposing it — but raise it early, because it is much harder to switch to once the campaign has started.

- **Track the campaign in the CSV, not in someone's head.** `SHAREDIPAD-AppleIDConflict.ps1` emits

tracking columns for a reason. A 40-person campaign with a tail of non-responders needs a list.

- **ABM is the truth; the script is the tracker.** Never present the script's candidate list to a

customer as a conflict list. It is not one, and saying it is will cost you credibility when the numbers disagree with ABM.

- **Give staff a real support path, not just a deadline.** The people you are asking to change their

Apple ID are field workers mid-shift. A deadline without help produces refusals.

Devices and procurement

- **Establish ABM device eligibility before you scope the project.** Serials not in ABM cannot be

ADE-enrolled and cannot run Shared iPad. On an inherited fleet bought piecemeal over years, this is frequently a procurement conversation, not a configuration one. Find out in the Assess phase.

- **Size storage for the resident-user count, and buy accordingly.** Under-specced iPads produce

sign-in failures and user eviction at exactly the moment the fleet goes live. Storage is cheap at purchase and expensive at wave 4.

- **Set the maximum cached users to what the hardware can actually carry** — not to the number of staff

on the roster. A device configured for more users than it can hold does not warn you; it fails.

- **Use VPP / ABM-licensed apps rather than store apps where you can.** A Shared iPad has no personal

App Store identity; device-licensed apps sidestep a whole class of install failure.

- **Pilot the CMMS app before you commit the fleet.** Ask the vendor, in writing, whether the app

supports Apple Shared iPad. If it does not honour the per-user boundary, you will find out at Test 5 — and no Intune setting will fix it.

Configuration

- **Use the allowlist, not a blocklist.** A blocklist of apps to hide is obsolete the moment a new app

exists. Name what is permitted and hide the rest.

- **Do not lock to a single app.** Kiosk single-app mode is right for an appliance and wrong here — the

field workload spans several apps and the user must be able to reach sign-out.

- **Block account modification.** It is the setting that stops a shift worker adding their personal

Apple ID or personal mail to a field iPad, and it is the one people forget.

- **Block predictive keyboard and autofill.** Shared keyboard state leaks identity across shifts. It is

a small setting and a real disclosure.

- **Keep Camera allowed.** It is part of the workload. Locking down the thing the field team actually

needs is how a deployment gets worked around.

- **Confirm Safari with the field team before blocking it.** If the CMMS relies on the browser for SSO or for opening links, blocking Safari breaks the workload — and the failure will look like a CMMS problem.

Naming and hygiene

- **Use the object names verbatim, everywhere.** `SHARED IPAD – FMD – Pilot`, `SHARED IPAD – FMD – Production`, `SHARED IPAD – ADE Profile – Shared`, `SHARED IPAD – Restrictions – Field Lockdown`, `SHARED IPAD – Compliance – iPadOS`, `SHARED IPAD – Apps – Field Allowlist`, `PILOT – Shared iPad Users`. The scripts match on exact display names, and the names use an **en dash (–)**, not a hyphen. A renamed object is an object the scripts cannot find and the next engineer cannot audit.

- **Keep Pilot and Production as separate groups permanently.** Do not collapse them after rollout. The next iPadOS change, the next app version, the next policy tweak all need somewhere safe to land first.

- **Use a company-owned Apple ID for the APNs certificate and record it.** A certificate created on a departed employee's personal Apple ID cannot be renewed, and the remedy is re-enrolling every managed device in the tenant.

Operations

- **Calendar both annual expiries the day you create them** — the APNs certificate and the ADE token. An expired APNs certificate breaks the entire fleet's management silently. An expired ADE token stops every new enrollment.

- **Verify the Provisioning Client group owner before every wave.** It is one click to check and a lost afternoon to diagnose. The failure — an empty group and nothing deploying — looks exactly like a policy problem.

- **Stage in waves of 8–10, never in one weekend.** A shift without devices is a shift not working. Validate each wave before starting the next.

- **Spot-check the session-isolation test on one device per wave.** Not just wave 1. Configuration drifts, and the test is cheap.

- **Brief the field team before their devices are staged.** Shared iPad gives a clean device *after* sign-out. A user who does not know to sign out leaves an open session on the rack — the one operational gap the architecture does not close for you. Print the handover runbook (Testing §5) and put it on the rack.

- **Designate a handover station with known-good Wi-Fi.** Sign-in and sign-out are network operations. A sign-out attempted in a dead spot in the yard is a sign-out that may not complete.

- **Keep the `SHAREDIPAD-Report.ps1` CSV as the as-built record**, and re-run it periodically. It is

your evidence that the fleet is still in the state you left it.

Scripts

• **Run everything read-only first.** Four of the five scripts cannot change anything. The fifth

defaults to `-WhatIf` and requires an explicit `-WhatIf:$false`. Read the dry-run output before you apply — that is what it is for.

• **Never present a script PASS as proof of session isolation.** `SHAREDIPAD-Validate.ps1` proves the

architecture exists. Only the A/B test with two real humans proves the sessions are isolated. That distinction is the difference between a delivered project and a callback.

• **Scope** `SHAREDIPAD-AppleIDConflict.ps1` **to a group or domain.** An unscoped tenant-wide worklist is

a list nobody can action.

8. Deliverables Summary — Shared iPad for Frontline Teams

Everything included in this solution package, and what each piece is for.

Documentation (this guide)

• **Executive Summary** — the business case in one page: the leak, the root cause, the A-vs-B decision,

the Managed Apple ID blocker, and the outcome.

• **Solution Overview** — the root cause stated plainly (why a shared single-user iPad *cannot* isolate

sessions), the two supported architectures with a side-by-side comparison and honest trade-offs, how this solution works, the Managed Apple ID reclamation workstream, benefits, limitations, the standard deployment model, and what the existing catalog already covers.

• **Requirements Guide** — 18 prerequisites, each with **Description · Why required · How to verify ·

How to implement if missing**: Intune and Microsoft 365 licensing, Entra ID, Apple Business Manager, verified domains,** devices present in ABM **(the constraint that decides feasibility on an existing fleet), the Apple MDM push certificate, the ADE token, supervision, iPadOS version,** storage sizing for per-user data**, Managed Apple IDs, ABM↔Entra federation,** the personal Apple ID conflict as a gate**, admin roles on both sides, network, Wi-Fi at the handover points, and executive sponsorship. Plus the category-level** Global Requirements — Intune Solutions**.

• **Implementation Guide** — exact, ordered, click-by-click steps across five phases: assess and prepare

identity (groups, the enrollment-time grouping owner, **the Apple ID conflict-resolution gate**, federation); configure the Apple connection (APNs, ADE token, **the enrollment profile with Shared iPad enabled** — the step that delivers isolation, device assignment); configure the device (field lockdown restrictions, compliance, the app allowlist with exact publisher/bundle IDs); pilot (wipe and re-enroll, validation gate); and expand (fleet eligibility, staged waves).

• **Testing & Validation** — nine tests with explicit PASS/FAIL gates, built around **the

session-isolation A/B test**: eight discrete checks (mailbox, cached mail, chat, photos, autofill, signed-in state, files, browser state) run with two real users on one device, in both directions, plus a diagnostic table mapping each failure back to the architecture step that caused it. Includes the** shift-handover runbook** for the field team.

• **Troubleshooting** — symptom → cause → fix, led by the #1 failure ("the next user still sees the

prior user's mail"), plus Managed Apple ID sign-in, ABM conflicts, federation loops, ADE enrollment, storage, app behaviour under Shared iPad, restrictions, and the scripts.

• **Best Practices** — architecture discipline, running the reclamation campaign without causing an

incident, procurement and sizing, configuration choices, naming hygiene, and operations.

PowerShell scripts (Microsoft Graph) — `SHAREDIPAD-Scripts.zip`

All scripts are tenant-side (run from an admin workstation with the Microsoft Graph PowerShell SDK). Each header states **Purpose · How to run · Required Graph scopes · Expected results**. **Four of the five make no changes of any kind. The fifth defaults to `-WhatIf`.**

1. `SHAREDIPAD-Discovery.ps1` *(read-only)* — readiness discovery: the ADE/ABM token and its expiry,

the Apple MDM push certificate and its expiry, every ADE enrollment profile **and whether Shared iPad is enabled on it**, the iPad inventory (model, storage, iPadOS, supervised, ADE state, ownership), and which of this solution's named objects exist.

1. `SHAREDIPAD-Validate.ps1` *(read-only)* — PASS/FAIL scoring against the success criteria: ADE

token valid, a **Shared-iPad-enabled** profile exists and is assigned, the device group is Assigned and owned by the Intune Provisioning Client, pilot devices supervised and ADE-enrolled, restrictions and compliance assigned, app allowlist assigned. Prints a clear PASS/FAIL summary — and says plainly that a PASS proves the *architecture*, not the isolation.

1. `SHAREDIPAD-Report.ps1` *(read-only)* — fleet report: per-device Shared iPad readiness with the

blocking reason, assigned ADE profile, supervision, storage, compliance, and last check-in, with CSV export. Use it to size the project and track waves.

1. `SHAREDIPAD-AppleIDConflict.ps1` *(read-only)* — builds the **Entra-side candidate worklist** of

users who need a Managed Apple ID and may therefore collide with a personal Apple ID, as a CSV with tracking columns for the reclamation campaign. **It states clearly that it cannot query Apple — the authoritative conflict list comes from Apple Business Manager; this is the candidate/tracking list only.**

1. `SHAREDIPAD-Set-Restrictions.ps1` *(the only state-changing script)* — assigns and verifies the

`SHARED IPAD – Restrictions – Field Lockdown` profile assignment, with a read-only pre-flight that checks the group is Assigned and owned by the Provisioning Client. **Defaults to `-WhatIf` and says so** — it changes nothing unless you pass `-WhatIf:$false` explicitly. It never removes an existing assignment.

Objects this solution creates

Object	Type
`PILOT – Shared iPad Users`	Entra security group (Assigned, users)
`SHARED IPAD – FMD – Pilot`	Entra security group (Assigned, devices)
`SHARED IPAD – FMD – Production`	Entra security group (Assigned, devices)
`SHARED IPAD – ADE Profile – Shared`	Intune ADE enrollment profile (Shared iPad enabled)
`SHARED IPAD – Restrictions – Field Lockdown`	Intune device configuration profile
`SHARED IPAD – Compliance – iPadOS`	Intune compliance policy
`SHARED IPAD – Apps – Field Allowlist`	Intune device configuration profile + the app assignments

Hero image

- `SHAREDIPAD-hero.png` — the product image (1200×1200) used on the listing.

How it's delivered

A **Word** document and **PDF** (cover, table of contents, page numbers, and a scripts appendix), plus all five PowerShell scripts in a single **ZIP** — yours to keep and reuse across **unlimited devices**.

A note on vendor specifics

This guide deliberately **flags rather than asserts** anything that Apple or Microsoft controls and changes: current Shared iPad prerequisites (iPadOS version, storage minimums, maximum resident users), Apple Business Manager portal navigation and flow names, the domain-verification and Apple ID conflict-resolution mechanics, ABM↔Entra federation specifics, and the Microsoft 365 Shared Device Mode iOS supported-app list. Each carries an explicit **verify against current Apple/Microsoft documentation** flag. The concepts, decision points, and sequencing are given confidently; the vendor-controlled specifics are pointed at the vendor's live documentation — because you are following this click-by-click, and a confidently-stated wrong number costs more than a flag.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

SHAREDIPAD-AppleIDConflict.ps1

Purpose: Every Shared iPad user needs a Managed Apple ID issued by Apple Business Manager. ABM CANNOT issue a Managed Apple ID at an address that is already in use by a PERSONAL Apple ID — so any staff member who once created a personal Apple ID with their work email address is a blocked Shared iPad user until that conflict is resolved (Requirements R14).

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once

SHAREDIPAD-Discovery.ps1

Purpose: Reports the tenant-side state of everything Apple Shared iPad depends on, so you know what is missing before you wipe a device: Apple MDM Push (APNs) certificate — present and expiry date. Apple ADE / Business Manager enrollment token(s) — present and expiry date. ADE enrollment profiles — and, for each, whether SHARED IPAD MODE IS ENABLED. iPad inventory — model, storage, iPadOS version, supervised state, ADE state, ownership. Which of this solution's named objects exist: SHARED IPAD - FMD - Pilot (Entra device group) SHARED IPAD - FMD - Production (Entra device group) SHARED IPAD - ADE Profile - Shared (ADE enrollment profile) SHARED IPAD - Restrictions - Field Lockdown (config profile) SHARED IPAD - Compliance - iPadOS (compliance policy) SHARED IPAD - Apps - Field Allowlist (config profile) PILOT - Shared iPad Users (Entra user group)

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\SHAREDIPAD-Discovery.ps1 .\SHAREDIPAD-Discovery.ps1 -CsvPath .\ipad-inventory.csv # also export the device inventory

SHAREDIPAD-Report.ps1

Purpose: Produces a per-device fleet report for the iPad estate and exports it to CSV. For each iPad: Device name, model, serial number, iPadOS version. Supervised state and enrollment type (ADE or not). Shared iPad READINESS — a computed verdict (READY / NOT READY) based on the device being ADE-enrolled and supervised, with the blocking reason when it is not. Assigned ADE enrollment profile (where the ADE token reports it) and whether that profile has Shared iPad enabled. Total and free storage — the constraint that bites at scale. Compliance state and last check-in.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\SHAREDIPAD-Report.ps1 .\SHAREDIPAD-Report.ps1 -CsvPath .\shared-ipad-fleet.csv .\SHAREDIPAD-Report.ps1 -DeviceGroupName 'SHARED IPAD – FMD – Production' -CsvPath .\wave3.csv

SHAREDIPAD-Set-Restrictions.ps1

Purpose: Verifies, and where missing adds, the group assignment of the configuration profile 'SHARED IPAD - Restrictions - Field Lockdown' to a named device group ('SHARED IPAD - FMD - Pilot' by default, 'SHARED IPAD - FMD - Production' at Expand).

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once

SHAREDIPAD-Validate.ps1

Purpose: Scores the tenant and the pilot devices against this solution's success criteria and prints a clear PASS/FAIL summary: 1. An Apple ADE token exists and has not expired. 2. The Apple MDM Push (APNs) certificate is active. 3. An enrollment profile named 'SHARED IPAD - ADE Profile - Shared' exists AND has SHARED IPAD ENABLED. 4. That profile is assigned to devices on the ADE token. 5. The pilot device group is Assigned (not Dynamic) and owned by the Intune Provisioning Client service principal (AppId f1346770-5b25-470b-88bd-d5744ab7952c) — without this owner the group stays empty at enrollment and nothing deploys. 6. Every pilot device is SUPERVISED and ADE-enrolled. 7. 'SHARED IPAD - Restrictions - Field Lockdown' exists and is assigned to the pilot group. 8. 'SHARED IPAD - Compliance - iPadOS' exists and is assigned to the pilot group. 9. 'SHARED IPAD - Apps - Field Allowlist' exists and is assigned to the pilot group.

How to run: Install-Module Microsoft.Graph -Scope CurrentUser # once .\SHAREDIPAD-Validate.ps1 .\SHAREDIPAD-Validate.ps1 -DeviceGroupName 'SHARED IPAD – FMD – Production'